Language as Visible Signal: Toward an Image-Native Language Model in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

Abstract

Most language models treat language as a sequence from a finite token and Unicode-accessible inventory, even when an image encoder or image generator is attached. That interface is incomplete for historical, regional, handwritten, damaged, undeciphered, and newly formed writing, and it discards visual form that can itself carry linguistic and historical evidence. OCR can transcribe the encodable subset but does not make writing the model's language substrate. We define a stricter target: an independent Imagized Language Model (ILM) whose inputs, learned states, predictive random variables, and primary outputs are continuous images of writing. Transcription is one supported task, alongside continuation, translation, explanation, comparison, and generation; it is not the architecture.

We contribute a staged, falsifiable implementation. Retinal Flow Language Models read ordered 32×32 fixations, integrate visual history, generate the next ink field, and reread their own pixels, but expose the failure of a monolithic language-and-typography writer. Predictive Visual Fields (PVF) factorize the problem: a convolutional retina maps writing to continuous state, a causal field integrates history, a deterministic proposal predicts the next state, an intrinsic hyperspherical flow models alternatives, and a separate motor planner renders pixels. The learned path has no tokenizer, strings after rasterization, Unicode or character IDs, OCR transcript, output vocabulary, discrete visual codebook, glyph lookup, candidate classifier, or external language model.

On a frozen 512-form, four-view Chinese benchmark, the 16.47M-parameter V16 proposal reaches 6.26% top-1, above last-image-only (3.97%), unigram (1.73%), and random dynamics (0.22%); full history adds +0.0773 normalized target log-probability. Its stochastic field reaches 3.69% versus 3.24% last-only. V16 trains on one RTX 4090 at 1.479 GiB peak allocated CUDA memory, but remains below a symbolic bigram (13.14%). A separate 5.73M-parameter V17 actuator proves causal state control on an untouched evaluation—58.59% visual identity versus 0.98% after shuffling intended state—but fails readability. A topology-first 2.36M-parameter V18 planner reaches 73.63% identity top-1 versus 0.98% shuffled and pixel F1 0.658 versus 0.313 on a fresh development audit at 0.778 GiB peak memory. Simple and medium forms become recognizable; dense forms remain malformed, and V18's frozen bank remains sealed. V19 then tests a zero-initialized spatial-retinal residual under fixed complexity strata and five causal interventions. Its fresh 512-candidate development audit rejects the mechanism: dense F1 is 0.728, but correct fields improve it by only 0.009 over shuffled fields and 0.005 over zero fields, against fixed margins of 0.12 and 0.03. V20 then constrains the global route to coarse 4×4 block logits and reserves within-block detail for a continuous local retinal field, with an exactly parameter-matched global-repeat control. The field becomes causally necessary: dense F1 is 0.701, versus 0.580 shuffled and 0.340 zeroed, and quadrant-occlusion locality is 1.0. The writer remains rejected because overall F1 is 0.636, target cosine is 0.815, the exact-decomposition tolerance fails, and final dense gain over the control is only 0.018 against a fixed

0.03 margin. Neither arm selects; no human or frozen query is authorized. V21 removes the remaining global spatial route: each local field cell emits coarse occupancy and 63 coefficients in an exact zero-DC patch basis, while global state supplies only spatially uniform modulation. At its best diagnostic snapshot, dense F1 is 0.705 versus 0.531 shuffled-field and 0.359 zero-field; identity top-1 is 79.10%, target cosine is 0.833, and locality and every algebraic invariant pass. The exact-parameter tiled-global control collapses to repeated patches. V21 still rejects the writer because overall, simple, and medium F1 (0.604, 0.565, and 0.594) miss fixed gates. No candidate selects, so the paired, human, and frozen stages remain forbidden. V22 then implements a six-frame visual prompt and one-frame image answer with two exactly matched 3.410M-parameter arms. A query-aware candidate sees the final visual query; a query-blind control receives a continuous null state. The candidate reaches only 0.78% counterfactual switch accuracy versus 0% control, 15.92% identity top-1 versus 15.33%, and pixel F1 0.511 versus 0.513. A development-only audit finds that its selector argmax is the operation frame in all 1,024 original and counterfactual prompts, while mean query attention is 1.37×10^{-13}. V22 therefore rejects single-frame selection as a relational binding mechanism. No candidate selects and frozen images remain uninstantiated. V23 replaces single-frame selection with relation before generation. A separately selected 1.122M-parameter image canonicalizer is frozen; a 25,602-parameter relation circuit compares query and label images, reads the operation image, routes one of two visible source glyphs, and emits a canonical answer image. On a fresh 1,024-episode development audit, the candidate reaches 99.80% query and operation switch accuracy and 99.51% identity top-1. Exact query-blind and operation-blind controls have zero switch and zero pixel change for their hidden factor. An opaque agent visual review scores 48/48 overall and 12/12 held-out. The single authorized frozen evaluation over 98 unseen identities reaches 99.83% binary choice, 99.61% query switch, 99.71% operation switch, 99.46% identity top-1, and 0.785 pixel F1. Every fixed gate passes without frozen model selection or threshold changes. This accepts a fixed two-pair same/other visual grammar, not free-form language. V24 removes absolute frame roles and fixed input length while closing a two-frame autoregressive visual loop. Visible header images locate roles in 15–24-frame packet streams; frame 1 renders the selected unseen glyph, and frame 2 is produced after rereading the actual generated frame-1 pixels. Only 1,347 parameters are trained. Fresh paired query-, operation-, history-, and header-blind controls localize every required cause. An opaque agent review scores 47/48 for frame 1 and 48/48 for frame 2. The single authorized frozen run over 107 unseen identities reaches 98.67% frame-1 identity, 99.73% frame-2 label accuracy, 97.66% query switch, 97.27% operation switch, and 99.61% history switch. Every fixed gate passes. This accepts a variable-input but fixed-grammar two-frame image stream, not arbitrary rendered language. V25 removes that grammar and trains a 25.55M-parameter causal field on ordinary Chinese book streams represented as 64 visible 32×32 cells. On 2,048 fixed development windows, full-history top-1 is 1.123%, versus 0.146% last-only, 0.342% after shuffling prior history, 1.611% image unigram, and 12.158% symbolic bigram. The ordered-history effect is measurable but six fixed semantic and causal gates fail. Peak allocated CUDA memory is 0.598 GiB, the student boundary passes, and the frozen partition remains sealed. An explicitly exploratory writer trained after rejection preserves ink density but obtains zero generated-identity top-1, reread cosine 0.080, and pixel F1 0.322. V25 therefore rejects this natural-language objective and writer rather than treating low memory or glyph-like marks as language efficiency.

The bounded product target is a Visual Word-Origin Book: rendered English or Chinese questions produce answer-page images containing readable explanation and provenance-linked historical or unencoded forms. The present experiments establish causal visual-language signal, compact continuous-image writing, a bounded image relation, and a bounded causal two-frame image stream, not autonomous question answering, broad readability, Qwen parity, or efficiency over text models. More generally, ILM treats writing as the first instance of language modeled as a continuous sensory field. A proposed Visual Language Stream adds sequence/time and optional geometric depth to the same image-native contract, so a page, book stream, 3D glyph string, or character movie differs by field shape rather than vocabulary. Only the 2D writing case is tested here; 3D, motion, speech, and vocal fields remain future hypotheses.

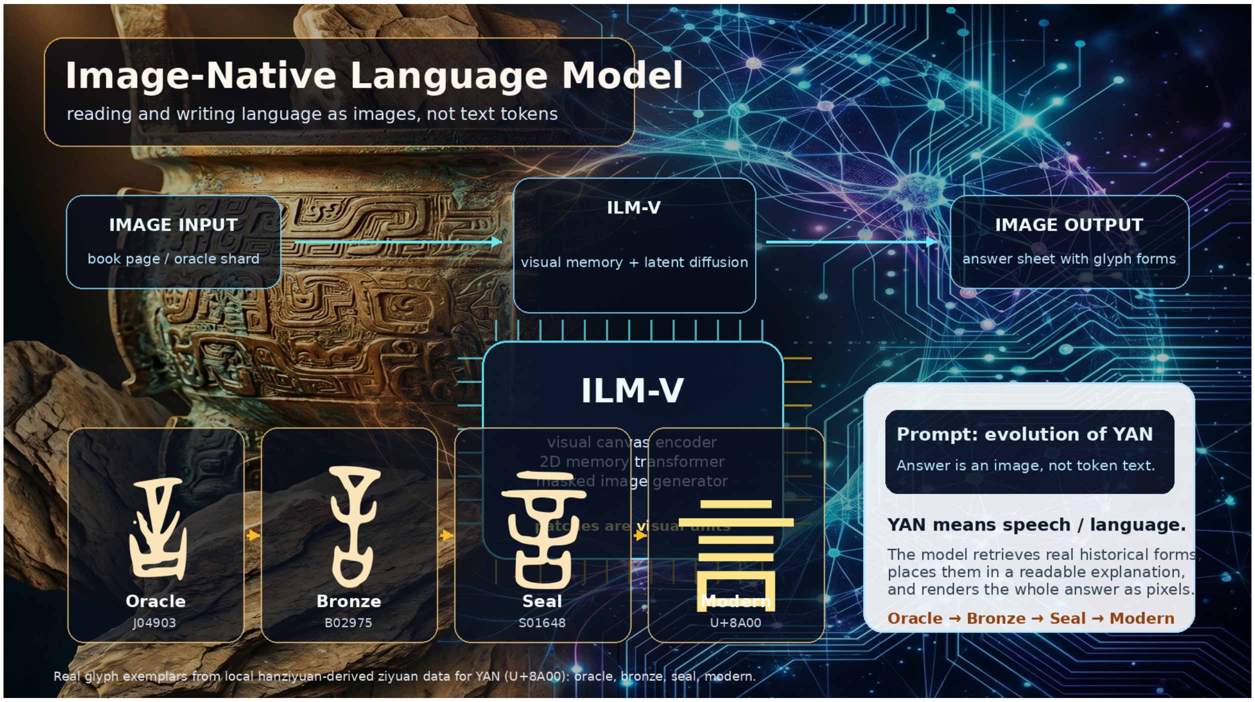

Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for YAN (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

1 Introduction: Writing as Language

1.1 Pain point

Human readers encounter language as marks on pages, screens, stone, bone, metal, and moving hands. Those marks vary with writer, era, medium, damage, typography, and reading direction. A finite token or Unicode interface is therefore not a generally accessible inventory of written language: it is an extremely useful codec for a documented subset. Historical and regional forms may have no code point, two visually different forms may share one code point, and an OCR transcript may erase the form whose evolution is the question.

Current language models are extremely capable, but their central language state is normally a sequence of symbolic tokens. Multimodal models can accept images, and modern APIs expose image generation [4, 5, 6]. These systems are important baselines, but attaching vision to a token-centered model does not make visible writing the predictive language substrate. An OCR system similarly solves a useful projection from image to available codes; it does not model all of the visual language that was projected away.

1.2 Goal and scope

The Imagized Language Model (ILM) instead treats written form as language:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output.}$$

The output should itself be a readable answer image. For a query such as “explain the evolution of character yan”, the system should produce a page containing modern explanation and attested oracle, bronze, seal, and modern forms, including forms unavailable in computer fonts. Typed input is rasterized before the core. A photographed page can be transcribed, but the same model should also continue, translate, explain, compare, and generate writing. Its answer is an image first; coded transcription is an optional downstream view.

The canonical input/output can be written as a Visual Language Stream,

$$X \in [0,1]^{T\times D\times H\times W\times C}, \qquad p_\theta(X_{t+1:t+k} \mid X_{\le t}),$$

where T is reading sequence or visual time, D is optional geometric depth, H, W are spatial extent, and C is the sensor channel. A flat prompt has $T = 1, D = 1$; an ordered line or book has $T > 1, D = 1$; a 3D glyph string has $D > 1$; and a 3D character movie uses depth and time. Collapsed axes have length one, so these are shapes of one continuous field rather than separate vocabularies. The present proof fixes $D = 1$ and short T; depth and motion must earn separate causal and quality evidence after the 2D loop passes.

Prompt following partitions this stream into observed prompt frames and generated answer frames,

$$X_{\text{prompt}} \in [0,1]^{T_p\times D\times H\times W\times C} \longrightarrow Y_{\text{answer}} \in [0,1]^{T_a\times D\times H\times W\times C}.$$

A typed question is deterministically rendered into prompt frames; a scan or inscription enters natively. $T_a = 1$ is an answer page and $T_a > 1$ is an ordered text-image stream or movie. A language claim requires correct held-out changes in the generated answer when the prompt meaning changes. Input reconstruction, OCR accuracy, glyph classification, or attractive writing alone is insufficient.

The design is inspired by foveated visual reading, without claiming a biological simulation. Its broader hypothesis is sensory: the same perception–memory– prediction–motor–self-perception loop might later operate on speech waveforms or time-frequency fields. This paper tests visible writing only.

1.3 Contributions

The contribution is not the slogan "text as pixels." It is a sequence of implemented controls that makes that slogan falsifiable:

1. A strict student boundary in which strings, token and Unicode IDs, OCR, finite visual codebooks, glyph lookup, and external language models are absent after rasterization and at deployed inference.

2. RFLM, a complete visual reading–prediction–writing–rereading loop that exposes autonomous generated pixels during evaluation and records its own negative result: stable ink is not readable language.

3. PVF, a mathematical factorization of continuous visual language state, causal memory, hyperspherical future-state uncertainty, and pixel actuation; V16 supplies controlled evidence that full visual history carries predictive language signal, while the symbolic bigram remains stronger.

4. Causal writer interventions: V17 proves that continuous image-derived intent controls generated identity, and V18 shows on development data that a 2.36M topology-first planner can render recognizable writing without token unembedding. Their failures delimit identity, topology, and language choice.

5. A preregistered V19 spatial-field intervention whose negative result separates feature availability from causal use, followed by an exactly capacity-matched V20 intervention that makes local topology necessary while rejecting its own writer-quality claim, and V21, which makes the field carry the complete spatial plan while identifying disjoint local raster patches as the next quality bottleneck.

6. V22, a strict six-frame image-prompt to image-answer experiment with a parameter-matched query-blind control. Its negative result localizes failure to relational binding: the selector collapses onto the operation image while writer quality and identity rise almost identically in both arms.

7. V23, a compositional visual relation circuit with an independently selected and frozen image canonicalizer, matched query-blind and operation-blind controls, held-out relation compositions, an opaque visual review, and one frozen-identity evaluation. It supplies the first accepted bounded image-prompt-to-image-answer result in this project.

8. V24, a variable 15–24-frame visual packet stream with learned header-image role localization and two autonomous output frames. Matched header-, query-, operation-, and generated-history-blind controls, an opaque visual audit, and one frozen run establish that frame 2 causally depends on rereading generated frame-1 pixels.

9. V25, a preregistered natural-Chinese visual-cell stream with fixed last-only, shuffled, blank, visual-frequency, text-bigram, and paired-history controls. It measures a weak ordered-history effect while rejecting the model under six fixed gates and preserving the frozen seal. A separately labeled writer diagnostic localizes stable ink without correct generated identity.

10. A rights-aware Visual Word-Origin Book benchmark and a continuous Visual Language Stream interface. They cover readable explanation, historical evidence, unencoded forms, optional depth and time, provenance, and one-GPU compute. The higher-dimensional interface is prospective.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. PIXEL reconstructs masked rendered-text patches and demonstrates cross-script visual transfer [10]; PIXAR is a genuinely generative pixel-space autoregressive language model and identifies noisy maximum-likelihood text images as a central difficulty [11]; PixelGPT studies autoregressive visual and textual pretraining [12]. These are the closest model-level precedents. Token-free language models such as CANINE, ByT5, and MEGABYTE remove subwords but retain symbolic bytes or characters. BLT and Scratchpad Patching allocate computation dynamically to high-entropy byte regions [16, 17]; ILM borrows the adaptive-compute question but not the byte substrate. Discrete diffusion language models show that generation need not be left-to-right, but usually retain a linguistic vocabulary.

Recent continuous approaches sharpen the distinction. Continuous visual autoregression can be trained with strictly proper energy scores without vector quantization [32]. VL-JEPA predicts continuous target-text embeddings but invokes a text decoder when output is required [33]. TextLDM reports that reconstruction-quality continuous latents are insufficient without alignment to a frozen pretrained language model [34]. These results motivate calibrated visual uncertainty and warn against equating reconstruction with semantics. Their byte, token, or text-decoder paths remain outside the independent ILM student boundary.

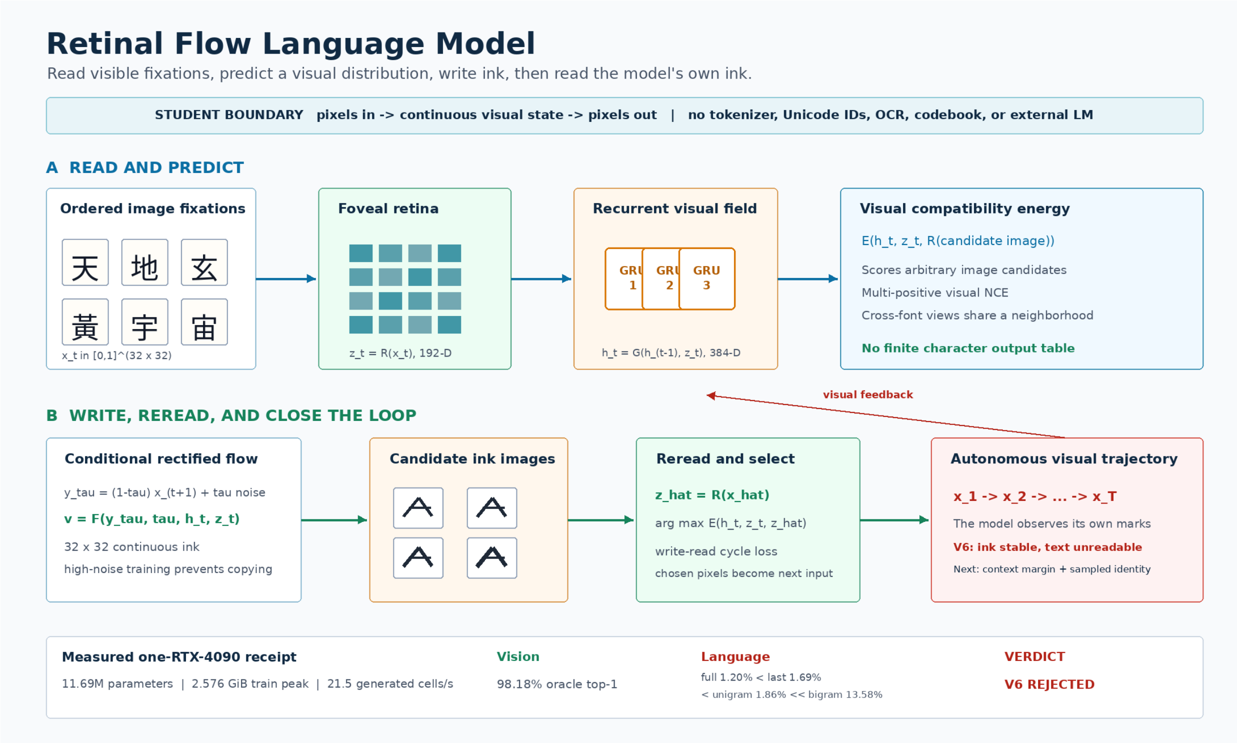

Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

I-JEPA shows that predicting target image representations from visual context can learn semantic visual fields without reconstructing every pixel [28]. Flow matching trains continuous generative paths by vector-field regression and supports efficient numerical sampling [25]. Human reading is also sequential and foveated: Chinese readers make saccades and can use both character- and word-related targeting cues [38]. These observations motivate RFLM, but they do not imply biological equivalence.

The remaining gap is not simply "pixels instead of tokens." It is a system whose predictive random variables are continuous writing images, whose memory is learned from ordered visual observations, whose writer produces new pixels, and whose own output distribution is present during evaluation. Historical evidence adds a separate constraint: synthesized marks must never be presented as attested source glyphs.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0,1]^{H\times W}$, RFLM factors a visual sequence as

$$p_\Theta(x_{1:T}) = p(x_1)\prod_{t=1}^{T-1} p_\Theta(x_{t+1} \mid x_{\le t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_\theta(x_t)/\|R_\theta(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar{\theta}}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_\phi(h_{t-1}, z_t), \qquad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_\psi(h_t, z_t, R_{\bar{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Conditional ink flow. For clean target x_{t+1}, noise ϵ, and $\tau \in [0, 1]$, define

$$y_\tau = (1 - \tau)x_{t+1} + \tau\epsilon, \qquad u_\tau = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_\omega = F_\omega(y_\tau, \tau, h_t, z_t), \qquad \mathcal{L}_{\text{flow}} = \|\hat{u}_\omega - u_\tau\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write–read cycle. The one-step endpoint $\hat{x}_0 = y_\tau - \tau\hat{u}_\omega$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\Big(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\Big).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg\max_{c \in \mathcal{C}_t} E_\psi(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

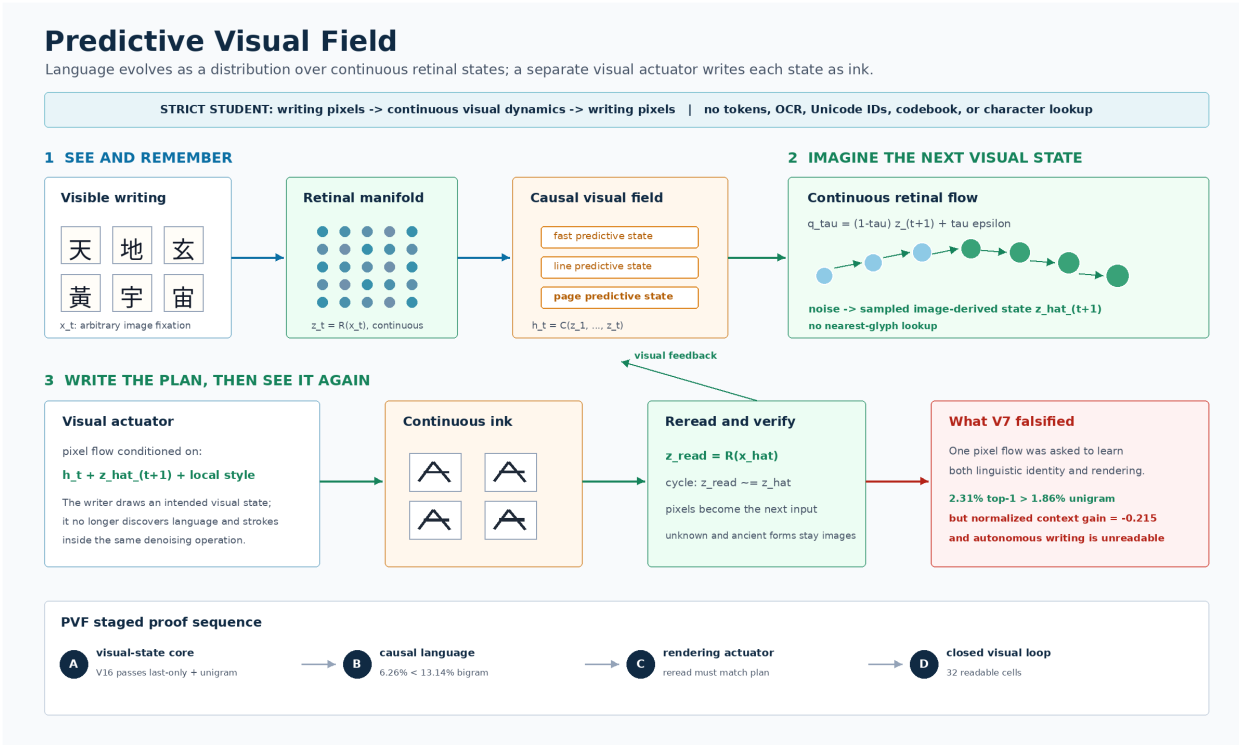

Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^{V}\right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_\psi([h_t, z_t])}{\|P_\psi([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source ϵ, let q_τ follow the shortest spherical geodesic from z_{t+1} to ϵ, and let u_τ be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E}\,\|v_\eta(q_\tau, \tau, [h_t, z_t]) - u_\tau\|_2^2 .$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \mathrm{Exp}_{q_k}\left(-\Delta\tau\, v_\eta(q_k, \tau_k, [h_t, z_t])\right),$$

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h^R_{1:t}$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory M_ϕ, the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g) W_o\, \mathrm{LN}\{M_\phi(h^R_{1:t}, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda\left[1 - \cos\{R_{\bar{\theta}}(\hat{x}_{t+1}), z_{t+1}\}\right].$$

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [31]. It combines the next-fixation representation objective of recurrent JEPA [29] with the prediction/generation separation explored by D-JEPA [30], and uses intrinsic flow matching on a Riemannian manifold [26, 27], while retaining an image-derived language state.

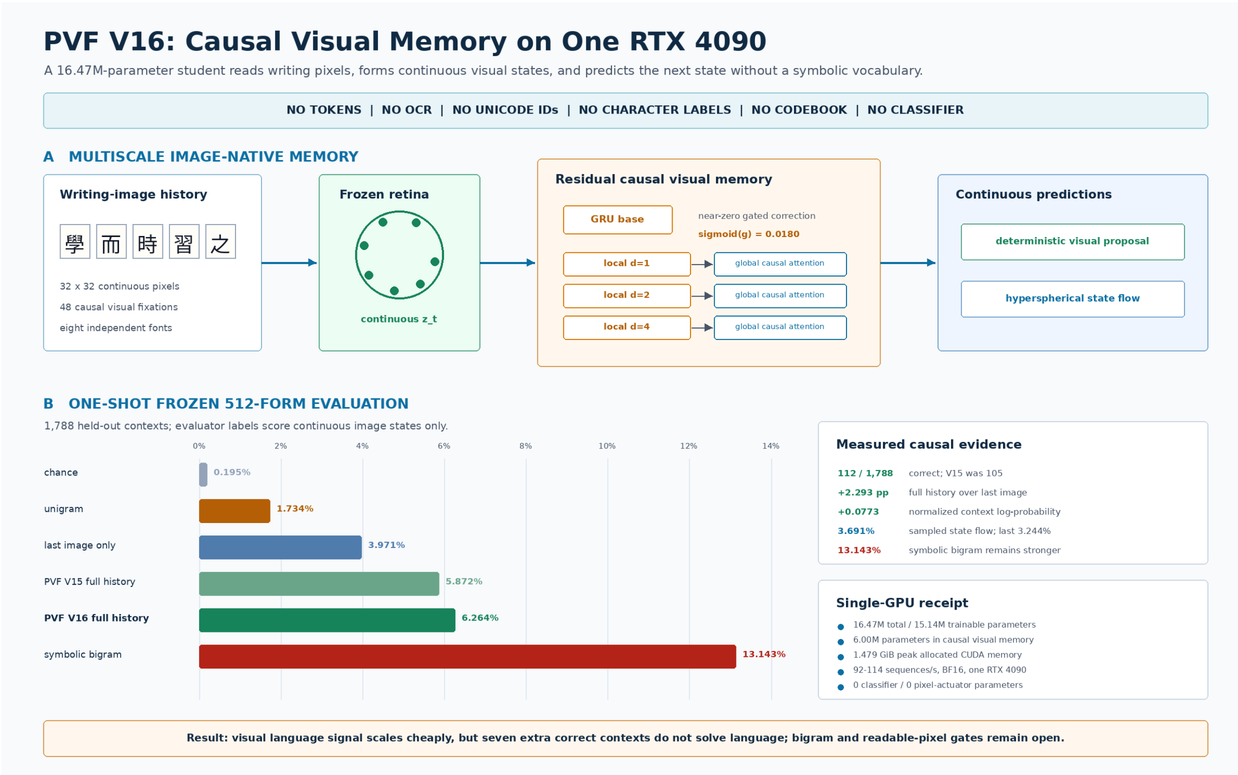

Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

5 Isolated Visual Actuation and Motor Planning

The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_\omega(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a 32×32 ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap.

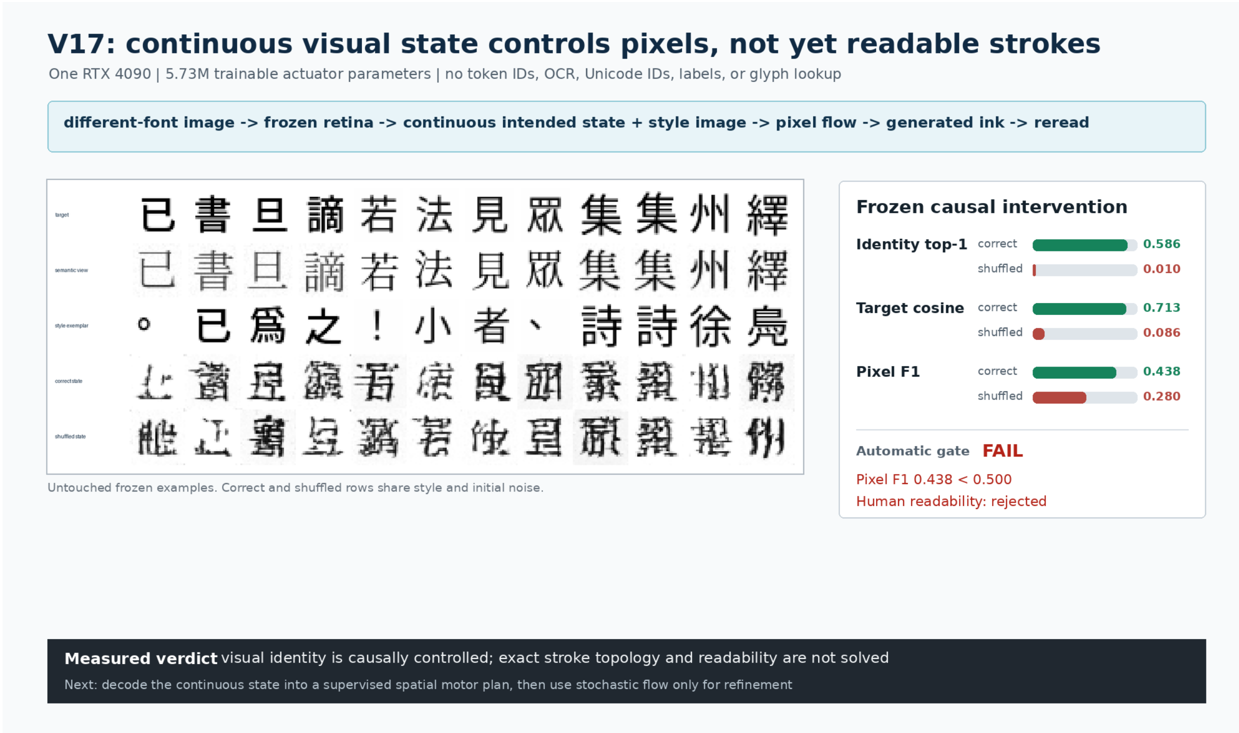

Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

5.1 Topology-first visual motor plan V18

V18 decodes state and style into a directly supervised spatial visual motor plan,

$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0,1]^{H\times W}.$$

A context MLP modulates a learned 4×4 seed, conditioned residual blocks upsample it to 32×32, and a sigmoid emits continuous ink. The loss joins stroke-weighted binary cross entropy, soft Dice, pixel and Sobel-edge distances, retinal identity and contrast, and a correct-versus-shuffled state margin. Target pixels supervise the output but never enter the condition. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook. Stochastic flow is reserved for later residual style and surface variation rather than stroke placement.

The salted partition contains 6,573 training, 223 development, and 221 frozen records. Selection maximized development correct-state pixel F1 subject to fixed topology, state-dependence, retinal-similarity, and non-copying gates. Step 1,400 was selected at pixel F1 0.6890 versus 0.3368 shuffled and global identity top-1 78.32% versus 1.17%. A fresh development seed yields 512 examples: identity top-1 is 73.63% versus 0.98%, target cosine is 0.8462 versus 0.0716, and pixel F1 is 0.6577 versus 0.3129. Mean ink fraction is 0.17505 in both branches and style-copy cosine is 0.0686.

Full-resolution review finds clear simple and medium forms but merged strokes in dense forms. Because the preregistration required human readability without defining a numeric blinded threshold, post-hoc inspection cannot authorize a frozen query. V18 therefore accepts readable topology emergence on development, rejects broad-complexity readability, and leaves the frozen identifier bank sealed. The next experiment must define complexity strata and a recognition rubric prospectively.

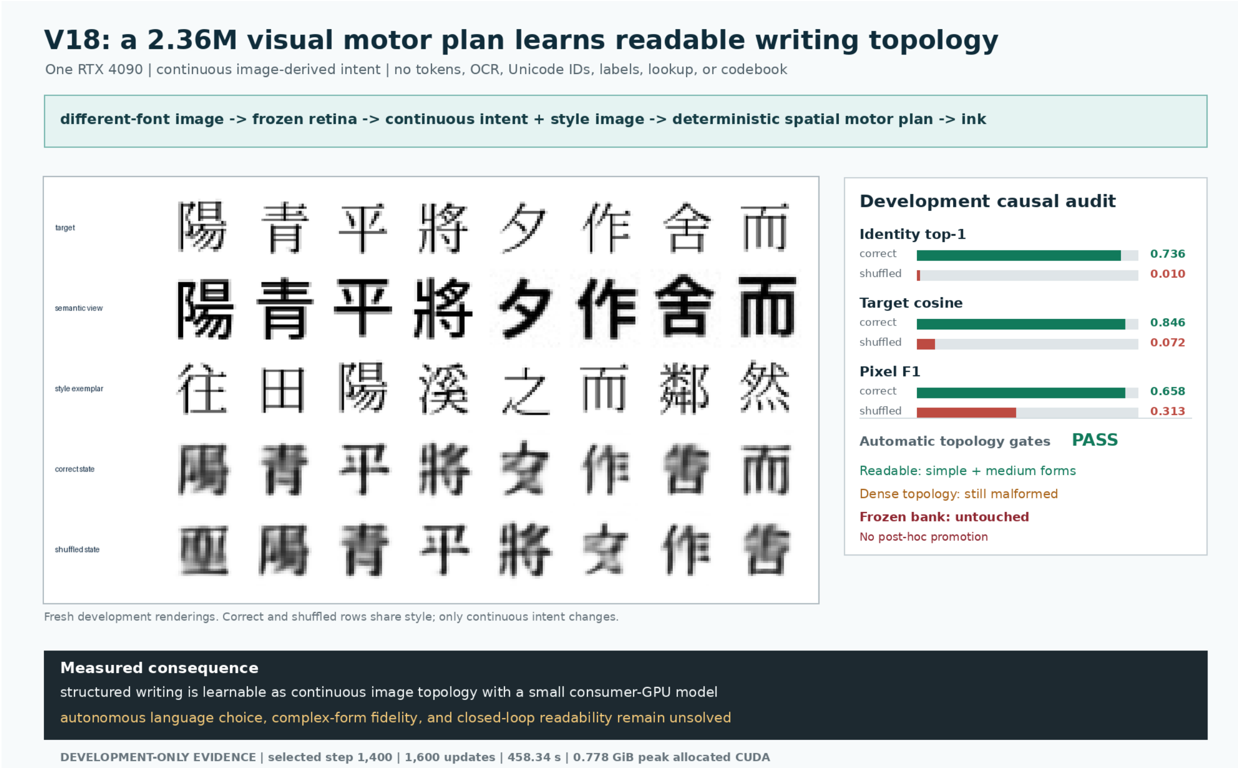

Figure 6: Measured V18 development result. A 2.36M-parameter deterministic motor planner turns continuous image-derived intent into recognizable writing. Shuffling only intended state destroys identity and topology while preserving mean ink occupancy. This is development-only evidence; the frozen bank remains untouched.

5.2 Spatial-retinal residual V19 and its rejection

V18 compresses the semantic image into one global vector before drawing. V19 tests the narrower hypothesis that this pooling discards component location in dense forms. The frozen retina now exposes both its existing global state and its pre-pooling spatial field,

$$(z, F) = R(x^{\text{semantic}}), \qquad z \in \mathbb{S}^{191}, \quad F \in \mathbb{R}^{192\times 4\times 4}.$$

Let $c = [z, E_s(x^{\text{style}})]$ and let $q_0 = S(c)$ be the learned global 4×4 seed of a V18-architecture planner. A conditioned convolutional adapter supplies a spatial residual,

$$a = A_\varphi(F, c), \qquad q = q_0 + \sigma(g)a, \qquad p = D(q, c).$$

The adapter's final projection is initialized exactly to zero, so V19 initially equals the selected global planner for every input even though its scalar gate is trainable. All global-planner and retina parameters remain frozen. This makes a gain attributable to spatial information rather than unrestricted baseline fine-tuning.

A clean causal comparison requires a new global baseline trained from scratch on the V19 partition. Loading V18 would contaminate V19 because V18's training split can intersect the new holdout. The selected clean global checkpoint is therefore frozen before the spatial adapter is trained; no V17 or V18 learned weight enters the experiment.

V19 development gate	Measured	Required	Result
Overall pixel F1	0.6710	> 0.68	fail
Dense pixel F1	0.7278	> 0.58	pass
Dense gain over shuffled field	0.0088	> 0.12	fail
Dense gain over zero field	0.0054	> 0.03	fail
Identity top-1	72.66%	$> 75\%$	fail
Target cosine	0.8416	> 0.84	pass

Table 1: Fresh 512-candidate V19 development audit. Four fixed automatic gates fail, so the blinded human review and frozen evaluation are not authorized.

Complexity is computed from target image geometry, never a character label. For thresholded ink b, define

$$C(b) = \bar{b} + \tfrac{1}{2}\left(\overline{|\Delta_x b|} + \overline{|\Delta_y b|}\right) + 0.1\,\overline{\mathbf{1}[\mathrm{AvgPool}_4(b) > 0.05]}.$$

The preregistered strata are simple $C < 0.24$, medium $0.24 \le C < 0.35$, and dense $C \ge 0.35$, weighted 1, 1.25, and 2 during spatial training. The objective is

$$\begin{aligned}\mathcal{L}_{19} = {} & \mathcal{L}_{\mathrm{WBCE}} + \mathcal{L}_{\mathrm{Dice}} + 0.5\mathcal{L}_1 + 0.25\mathcal{L}_{\mathrm{edge}} \\ & + 0.05\mathcal{L}_{\mathrm{id}} + 0.05\mathcal{L}_{\mathrm{NCE}} + 0.1\mathcal{L}_{\text{spatial-margin}} + 0.1\mathcal{L}_{\text{zero-margin}}.\end{aligned}$$

Rowwise terms use the complexity weights. The margins require correct fields to outperform batch-shuffled fields by 0.03 and the zero-field branch by 0.01 in target pixel error.

Evaluation fixes target and style while comparing five branches: correct global state plus correct field; correct global plus shuffled field; shuffled global plus correct field; both shuffled; and correct global plus zero field. The prospective gate requires overall pixel F1 above 0.68, dense F1 above 0.58, dense spatial-shuffle and zero-field advantages, global identity above 0.75, and 38/48 blinded recognitions with stratum-specific minima. The implementation passes exact-equivalence, gradient-isolation, field-shape, intervention, complexity, and serialization tests.

A clean global planner was trained from scratch for 1,600 updates on the V19 partition, selected at step 1,600, and frozen. It reached development F1 0.6754, identity top-1 77.93%, and target cosine 0.8339. The V19 intervention then trained only 764,545 adapter parameters for 1,600 updates in 329.51 seconds, using 0.899 GiB peak allocated CUDA memory on one RTX 4090. A separate deterministic audit rendered 512 new development candidates. Table 1 records the fixed decision.

This result rejects the additive routing mechanism, not the possibility that the field contains useful information. The global planner can already draw the whole form, so ordinary reconstruction gradients need not use the field. The next bounded candidate must make local visual state structurally necessary.

5.3 Retinal topology router V20: local causality without writer selection

V20 implements the preregistered coarse/detail decomposition

$$p = \sigma\left(U(G(z, s)) + (I - UD)H(F, s)\right),$$

where G emits only one logit per 8×8 output block and H maps each 4×4 retinal-field cell pointwise to its corresponding 8×8 detail patch. The fixed projection $I - UD$ removes each patch mean, so the global route cannot represent within-block detail. An exactly capacity-matched control replaces F with $\mathrm{tile}(z)$ while retaining the same local decoder, losses, and operations. Each arm has 506,448 trainable parameters.

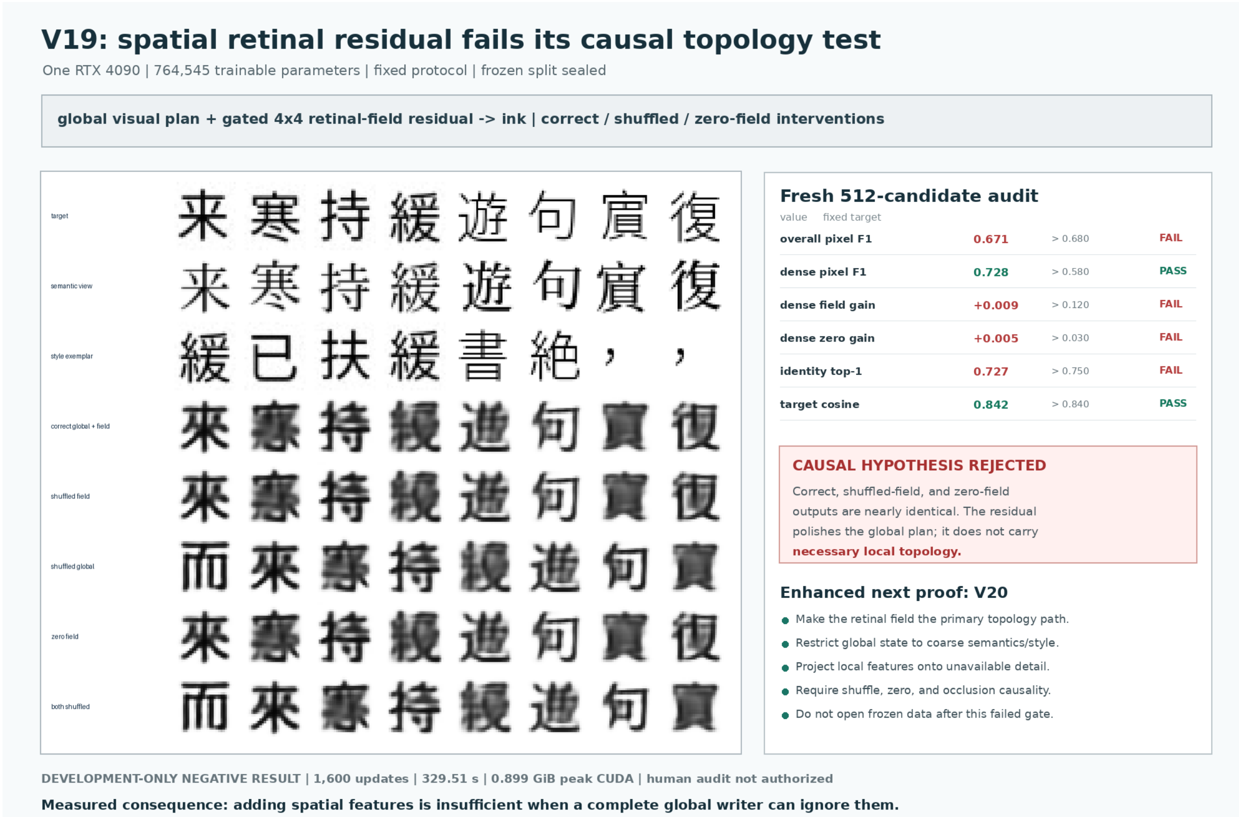

Figure 7: Measured V19 result. Correct, shuffled-field, and zero-field rows are nearly identical, while shuffling the global state changes identity. The spatial adapter learned a shared residual polish rather than necessary local topology.

The new salted split contains 6,611 training, 196 development, and 210 frozen records. Both arms train for 1,600 updates against the same frozen V16 retina. The fixed interventions hold target and style constant while shuffling the local field, zeroing it, shuffling global state, shuffling both, or occluding each field quadrant. Validation renders 512 development candidates, including 146 dense forms. Table 2 records the final endpoints; neither is a selected checkpoint.

The candidate's correct dense F1 exceeds shuffled-field by 0.12177 and zero-field by 0.36128; its quadrant locality is exactly 1.0. These measurements establish that the continuous field carries necessary, topographically aligned detail. They do not establish a selected writer. Overall F1 and reread cosine remain below threshold. Both arms also exceed the strict 10^{-6} detail-mean tolerance after detail magnitude grows, indicating that floating-point recentering is a poor way to encode an algebraic invariant. Candidate dense F1 exceeds the control by only 0.01791 at the final endpoint, below the paired 0.03 margin. Candidate and control train in 325.90 and 327.11 seconds and peak at 0.323 and 0.397 GiB allocated CUDA memory, respectively.

5.4 Field-complete writer V21: complete local route, rejected rasterizer

V21 tests the field-complete correction rather than widening V20. Global state and image-derived style emit only spatially uniform modulation. Each local field cell independently emits one coarse

V20 development gate	Candidate	Required	Result
Overall pixel F1	0.6361	> 0.66	fail
Dense pixel F1	0.7013	> 0.68	pass
Dense gain over shuffled field	0.1218	> 0.12	pass
Dense gain over zero field	0.3613	> 0.10	pass
Identity top-1	75.00%	$> 70\%$	pass
Target cosine	0.8152	> 0.82	fail
Occlusion locality	1.0000	> 0.40	pass
Detail block-mean magnitude	2.03×10^{-6}	$< 10^{-6}$	fail
Dense gain over matched control	0.0179	> 0.03	fail

Table 2: Final V20 development endpoints. The field candidate passes every local-causality gate but fails writer selection and the matched-control margin. Because neither arm selects, the formal paired audit, blinded review, and frozen evaluation remain forbidden.

V21 candidate development gate	Measured	Required	Result
Overall pixel F1	0.6038	> 0.66	fail
Simple pixel F1	0.5648	> 0.58	fail
Medium pixel F1	0.5945	> 0.60	fail
Dense pixel F1	0.7053	> 0.70	pass
Dense gain over shuffled field	0.1739	> 0.15	pass
Dense gain over zero field	0.3465	> 0.20	pass
Identity top-1	79.10%	$> 74\%$	pass
Target cosine	0.8331	> 0.82	pass
Occlusion pixel change	0.0926	> 0.03	pass
Occlusion locality	1.0000	> 0.95	pass
Detail block-mean magnitude	3.87×10^{-7}	$< 5\times10^{-6}$	pass
Decomposition error	0.0	$< 10^{-6}$	pass

Table 3: Best V21 candidate diagnostic snapshot. It passes every causal, identity, reread, and algebraic gate but fails overall, simple, and medium pixel fidelity. It is not a selected checkpoint.

coefficient and 63 detail coefficients,

$$(\gamma, \beta) = M(z, s), \qquad (c_{ij}, a_{ij}) = A_F(\gamma \odot F_{ij} + \beta), \qquad \ell_{ij} = c_{ij}\mathbf{1} + B_0 a_{ij},$$

where B_0 contains the 63 non-DC columns of a normalized Walsh–Hadamard basis. Its column sums and Gram error are exactly zero in the recorded FP32 computation. The 16 nonoverlapping 8×8 patches make the 32×32 logit field. There is no coordinate input, position parameter, cell mixing, global spatial projection, or learned glyph table. The matched control tiles z into all 16 source cells while retaining every parameter and operation. Each arm has 582,336 trainable parameters.

The preregistered salted split has 6,613 training, 195 development, and 209 frozen records. Both arms train for exactly 1,600 updates. No candidate checkpoint passes every selection gate. Table 3 therefore reports step 1,400 only as the best diagnostic snapshot under the fixed ranking, not as a selected model.

The tiled-global control selects its structural step 1,200 and reaches only 0.1473 overall F1, 0.3074 dense F1, 25.00% identity top-1, and 0.4332 target cosine. Candidate step 1,400 exceeds those values by 0.4564 overall F1 and 0.3979 dense F1. This comparison is descriptive rather than the formal paired audit: the snapshots use different development draws and steps, and the paired

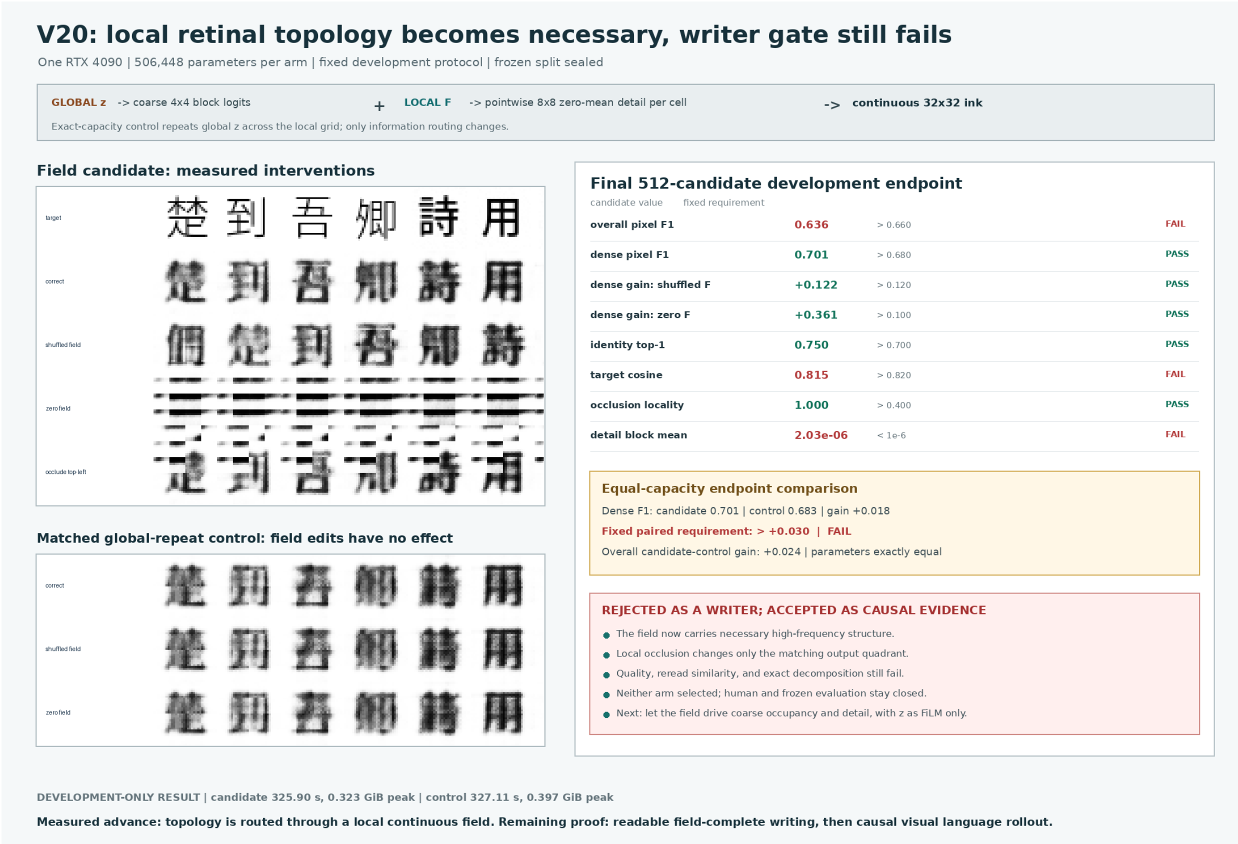

Figure 8: Measured V20 result. Shuffling or zeroing the candidate's local field destroys detail, and a quadrant occlusion changes only its corresponding output quadrant. The equal-parameter global-repeat control is invariant to field interventions. Recognizable endpoints do not pass the fixed writer and paired-control gates.

evaluator requires a selected candidate before constructing a shared audit. It must therefore refuse V21.

Candidate and control train in 327.04 and 333.41 seconds at 0.325 and 0.400 GiB peak allocated CUDA memory. Frozen-image count remains zero. V21 therefore establishes complete, necessary, topographic use of continuous local state, but rejects nonoverlapping pointwise patch emission as a broadly readable writer. The next writer test should add overlapping local support with a fixed partition-of-unity blend, or an equivalently bounded multiscale local operator, while retaining an exact-parameter tiled-global control and a measured influence radius.

5.5 Visual Binding Stream V22: image prompt, rejected relation

V22 removes supplied semantic intent and asks the smallest prompt-following question that also requires arbitrary visible form. Every episode is a six-frame image stream,

$$X = (l_1, g_1, l_2, g_2, o, q),$$

where two visible labels bind two distinct, previously unseen glyph images. The operation image is either the visible form for same or other, and the final query is one of the two label images.

Figure 9: Measured V21 result. The candidate uses local state for complete spatial structure: shuffling or zeroing the field corrupts the form and a local occlusion stays local. The equal-parameter tiled-global source repeats one patch. Disjoint candidate patches remain blurred and seamed, causing the preregistered broad-fidelity rejection.

A paired counterfactual changes only q; the correct canonical answer image must switch. Source glyphs use independent noncanonical faces, so exact pixel copying cannot satisfy the target.

The frozen V16 retina maps each frame to global and local continuous states. A four-block visual Transformer forms a query from its final hidden state and a softmax over all six frames. Weighted global/local values feed an overlapping 12×12, stride-8 local writer with a fixed positive partition-of-unity blend. The query-blind control replaces only the final query retinal state with a learned continuous null state. Both arms contain exactly 3,410,128 trainable parameters with identical shapes. The student path contains no string, token or Unicode ID, OCR, character/operation label, answer index, glyph lookup, finite codebook, or external language model.

The fixed identity bank contains 815 training, 104 development, and 105 frozen Chinese forms under a salted split; two operation/label combinations are held out compositionally. Each arm trains for exactly 1,600 updates. Table 4 reports the final development endpoints. Candidate and control writer/identity trajectories nearly overlap, while prompt switching remains absent.

The mechanism audit makes the rejection more specific than failure to converge. Candidate mean attention is 1.0 on the operation frame, 2.49×10^{-12} and 5.53×10^{-12} on the two glyph frames, and 1.37×10^{-13} on the query. The entropy penalty rewards this confident shortcut. The task, however, is a relation among operation, query, labels, and glyphs; no single frame contains the answer. Candidate training takes 258.70 seconds at 0.315 GiB peak allocated CUDA memory, so

V22 candidate development gate	Measured	Required	Result
Binary choice	0.4824	> 0.85	fail
Counterfactual switch	0.0078	> 0.80	fail
Held-out-combination switch	0.0113	> 0.75	fail
Identity top-1	0.1592	> 0.45	fail
Identity gain over query shuffle	0.0225	> 0.20	fail
Target cosine	0.4601	> 0.78	fail
Pixel F1	0.5107	> 0.58	fail
Oracle-writer F1	0.6020	> 0.64	fail
Paired output L1	0.0089	> 0.08	fail
Frozen images instantiated	0	0	pass

Table 4: V22 query-aware endpoint. The anti-copy margins and structural receipts pass, but every prompt-dependence and output-quality gate fails. The query-blind control ends at 0 switch accuracy, 0.1533 identity top-1, 0.5125 pixel F1, and exactly zero paired-output difference.

model/GPU capacity is not the identified bottleneck.

No candidate checkpoint selects. The paired evaluator refuses before fresh data construction, blinded review is unauthorized, and no frozen image is instantiated. V22 establishes a real image-prompt to image-answer interface and weak unseen-form writing only; it does not establish prompt understanding.

V23 removes unrestricted single-frame selection. A shared visual comparator must match query and label images, while an operation-image reader provides a continuous gate,

$$a_i = C_\theta(R(q), R(l_i)), \quad m_i = \mathrm{softmax}_{i\in\{1,2\}}(a_i), \quad s = \sigma(U_\theta(R(o))),$$

$$w_i = sm_i + (1-s)(1-m_i), \quad (\hat{z}, \hat{F}) = \sum_{i=1}^{2} w_i V(R(g_i)).$$

This relation circuit composes multiple visible roles before writing. Its operation semantics and equality function are learned from answer images; no symbolic target index enters the student. Exact query-blind, operation-blind, pair-swap, query-counterfactual, and operation-counterfactual interventions are required before any longer answer stream. A qualified writer should be frozen first so binder/writer co-adaptation cannot hide another average-glyph route.

5.6 Visual Relation Circuit V23: bounded prompt relation accepted

V23 fixes the V22 causal defect while preserving the same strict visual interface. Its bank contains 817 training, 109 development, and 98 frozen Chinese identities under the preregistered salt `visual-relation-circuit-v23`. The two held-out operation/label compositions never occur during training, although every individual operation and label pair does. Every model call receives only floating image tensors or continuous states derived from those tensors; renderer strings and episode metadata are deleted before `forward`.

Stage A independently trains a 1,122,081-parameter convolutional U-Net to map a noncanonical source-glyph image to a canonical target image. It has no identity embedding, target-font ID, coordinate channel, or character table. The selected step-1,000 checkpoint reaches 0.7583 pixel F1, 0.9951 identity top-1, and 0.9375 target cosine on all 109 unseen development identities. Its pixel-F1 gains over the raw source and a shuffled source are 0.2173 and 0.4237. The canonicalizer is then frozen before relation training.

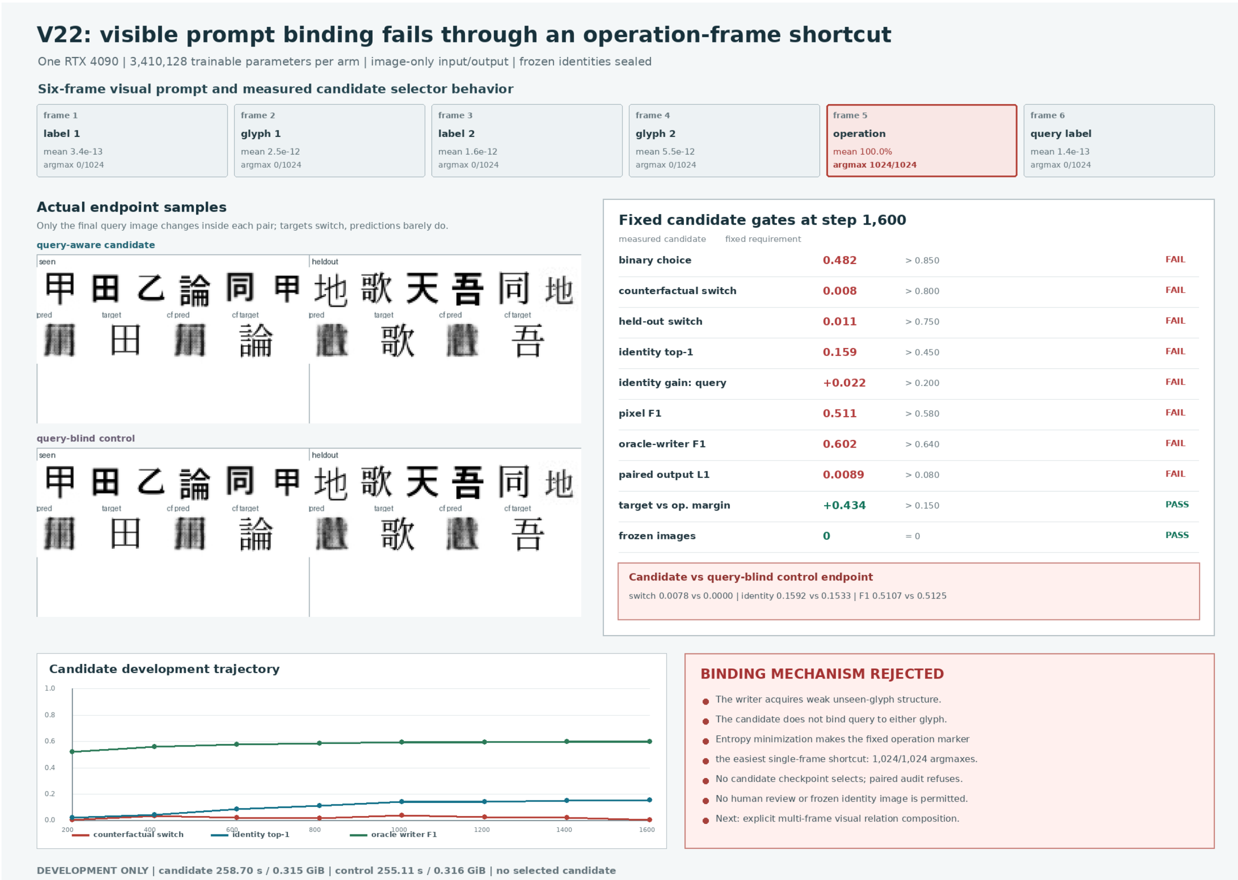

Figure 10: Measured V22 result. Actual endpoint samples show that targets switch when only the final visual query changes, while candidate predictions barely move and match the query-blind control. The candidate selector assigns every one of 1,024 development argmaxes to the operation frame. Writer metrics improve without relational prompt binding.

Stage B uses normalized frozen-retina states. For each label image,

$$c_i = \langle R(q), R(l_i) \rangle, \qquad m_i = \mathrm{softmax}_{i \in \{1,2\}}(\tau c_i),$$

where the positive temperature τ is learned. A small operation-image reader produces

$$s = \sigma(U_\theta(R(o))),$$

and composes match and operation before generation:

$$w_i = s m_i + (1-s)(1-m_i), \qquad x_r = w_1 g_1 + w_2 g_2, \qquad \hat{y} = C_\psi(x_r).$$

Thus same routes the visually matched source and other routes its complement. The operation semantics are learned from answer-image loss, not supplied as a class label. The routed variable is a continuous source-pixel field, and C_ψ is the frozen Stage A canonicalizer. The trainable relation circuit has 25,602 parameters: the operation reader, one temperature scalar, and two continuous null states.

Candidate, query-blind, and operation-blind arms have exactly identical parameter names and shapes. Query-blind replaces only $R(q)$ with its learned continuous null state; operation-blind

Fresh paired metric	Relation-aware	Query-blind	Operation-blind
Binary choice	0.9990	0.5010	0.5002
Query switch	0.9980	0.0000	0.0615
Operation switch	0.9980	0.0156	0.0000
Identity top-1	0.9951	0.2947	0.2993
Pixel F1	0.7563	0.5407	0.5404
Query-output L1	0.1915	0.0000	0.0109
Operation-output L1	0.1912	0.0017	0.0000
Pair-swap output L1	0.0000	0.0000	0.0000

Table 5: V23 fresh paired development audit. A control has exactly zero response to its blinded factor by construction, while the relation-aware arm retains both dependencies with matched parameter shapes.

V23 frozen gate	Measured	Required	Result
Binary choice	0.9983	> 0.95	pass
Query switch	0.9961	> 0.90	pass
Operation switch	0.9971	> 0.90	pass
Held-out minimum switch	0.9961	> 0.85	pass
Identity top-1	0.9946	> 0.75	pass
Pixel F1	0.7848	> 0.68	pass
Target cosine	0.9399	> 0.82	pass
Query-label visual match	0.9995	> 0.98	pass
Operation-gate accuracy	1.0000	> 0.98	pass
Pair-swap consistency	1.0000	> 0.99	pass

Table 6: Single authorized V23 frozen evaluation over previously unseen identities. Every threshold was fixed before implementation and remained unchanged.

replaces only $R(o)$. All arms process base, query-counterfactual, operation-counterfactual, and pair-swapped image prompts jointly for 600 fixed updates. The selected candidate reaches 0.9946 binary choice, 0.9902 query switch, 0.9902 operation switch, 0.9873 identity top-1, and 0.7548 pixel F1 on checkpoint-selection development data.

A fresh paired audit then constructs 1,024 new development episodes and 4,096 prompt variants. Table 5 reports the exact interventions. The candidate changes identity and pixels when either visible causal factor changes. Each blind control is exactly invariant to the factor it cannot see, and all arms remain exactly invariant to pair order.

After the paired gate passes, an opaque development review presents 48 image cards without transcriptions, targets, correctness labels, or frozen examples. The reviewing agent chooses which of the two visible source forms the generated answer depicts before the scorer opens the sealed key. It obtains 48/48 overall and 12/12 on held-out compositions, above the fixed 44/48 and 11/12 gates. This is an agent visual review, not a human-subject study.

Only then does the frozen evaluator instantiate the 98 held-out identities. It runs once on 1,024 episodes, 4,096 prompt variants, and four identity-bank views, with no model selection or threshold changes. Every gate in Table 6 passes. The run takes 4.74 seconds and peaks at 0.498 GiB allocated CUDA memory. Stage A training takes 214.10 seconds at 0.700 GiB; candidate relation training takes 117.18 seconds at 1.406 GiB, all on one RTX 4090.

V23 therefore accepts a bounded image-only relation: multiple visible roles are composed, unseen source forms are routed, and the answer is emitted as pixels. It does not establish arbitrary

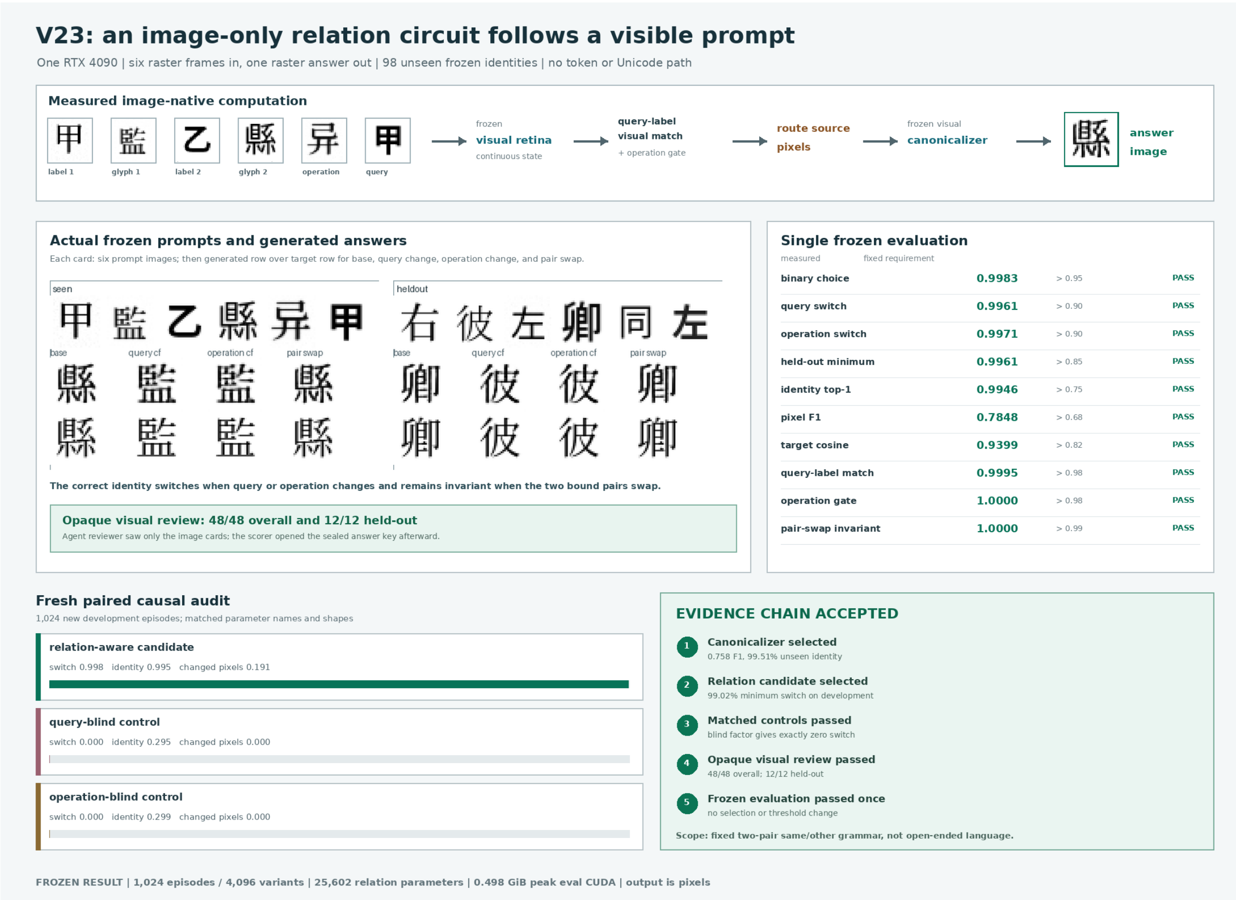

Figure 11: Measured V23 result. Actual frozen samples show generated answers above their targets for base, query-counterfactual, operation-counterfactual, and pair-swapped prompts. The matched blind controls localize both causal dependencies, and the single frozen run passes every fixed gate.

syntax or free-form generation. Frame roles and the two-pair same/other algebra remain architecturally fixed, and the answer is a canonicalized version of one visible source form. V24 implements the preregistered next test: remove absolute frame roles, consume a variable-length visual instruction, emit two answer-image frames, and make frame 2 causally depend on rereading the generated pixels of frame 1.

5.7 Visual Packet Reread Stream V24: causal two-frame stream accepted

V24 replaces V23's six absolute roles with a variable sequence of three-frame visual packets. Its prompt and autonomous answer have shapes

$$X \in [0,1]^{B\times T\times 1\times 32\times 32}, \quad T \in \{15, 18, 21, 24\}, \qquad \hat{Y} \in [0,1]^{B\times 2\times 1\times 32\times 32}.$$

Each packet is *header, content A, content B*. The visible header glyphs for pair (U+5951), operation (U+6CD5), query (U+95EE), distractor (U+65C1), and stop (U+6B62) denote their respective roles. Packets before the terminating packet are shuffled. Training uses 15, 18, and 21 active frames; 24 frames with three distractors are held out by length. Dynamic batching adds only all-zero packet images and supplies neither active lengths nor padding masks.

The identity bank contains 829 training, 88 development, and 107 frozen Chinese forms under the fixed salt `visual-packet-reread-stream-v24`. Two operation/label-pair compositions are held out while every constituent operation and label pair appears separately in training. As in V23, renderer strings and episode metadata are deleted before every model call.

For packet j, the frozen V16 retina yields normalized header and content states H_j, A_j, B_j. Three learned normalized prototypes locate roles from header pixels:

$$e_{jr} = \tau_r \langle H_j, p_r \rangle, \qquad r \in \{\text{pair}, \text{operation}, \text{query}\}.$$

Straight-through hard routing selects the top two pair packets and top one operation and query packet; its backward surrogate is a softmax. Scores inspect only header images, not packet contents. With located query q, operation o, pair labels A_j, and pair-glyph images B_j^{image}, frame 1 uses the frozen V23 comparison temperature, operation reader, and canonicalizer:

$$m_j = \text{softmax}_{j \in \text{pair}} \left(\tau_{23} \langle q, A_j \rangle \right), \qquad s = \sigma(U_{23}(o)),$$

$$w_j = s m_j + (1 - s)(1 - m_j), \qquad x_1 = \sum_{j \in \text{pair}} w_j B_j^{\text{image}}, \qquad \hat{y}_1 = C_{23}(x_1).$$

The second step rereads the actual generated pixels:

$$r_1 = R(\hat{y}_1), \qquad b_j = \text{softmax}_{j \in \text{pair}} \left(\tau_{23} \langle r_1, B_j \rangle \right),$$

$$x_2 = \sum_{j \in \text{pair}} b_j A_j^{\text{image}}, \qquad \hat{y}_2 = C_{23}(x_2).$$

Thus no pre-render identity state crosses the frame-1/frame-2 boundary. All weighted reductions accumulate in FP64.

V16 retina parameters and every inherited V23 parameter are frozen. V24 trains only three 192-dimensional role prototypes, four 192-dimensional continuous null states, and three role temperatures: 1,347 parameters total. The packet-aware, header-blind, query-blind, operation-blind, and history-blind arms have identical parameter names, shapes, and counts. Each receives exactly 800 AdamW updates in BF16 with batch 64 and eight workers on one RTX 4090. The candidate selects step 100, then completes the fixed run in 394.18 seconds at 2.322 GiB peak allocated CUDA memory.

A fresh audit uses 1,024 newly rendered development episodes at seed 22260832. Table 7 shows the paired causes. A blind arm is exactly invariant to the intervention it cannot observe. Header blindness prevents reliable role localization, while history blindness leaves frame 1 intact but breaks frame 2.

After every paired gate passes, an opaque agent visual audit presents 48 development cards containing only the packet images, two visible source choices, and two generated frames. It exposes no targets, transcriptions, correctness labels, or frozen forms. The committed judgments score 47/48 for frame 1 and 48/48 for frame 2, including 12/12 for each frame at held-out $T = 24$. This is an agent visual audit, not a human-subject study.

Only then does the evaluator instantiate frozen images. It runs exactly once on 1,024 episodes, 107 unseen identities, and four identity-bank views, without model selection, threshold changes, or repetition. Table 8 reports the primary gates. All remaining role, distractor, packet-permutation, teacher-forcing, and image-boundary gates also pass. The run takes 8.13 seconds and peaks at 0.542 GiB allocated CUDA memory.

V24 therefore accepts a variable-input, two-frame image stream under one fixed visual grammar. It establishes that frame 2 can depend causally on rereading generated frame-1 pixels, not merely on

Fresh paired metric	Packet-aware	Corresponding blind
Query switch	0.9922	0.0000
Query-output L1	0.1677	0.0000
Operation switch	0.9932	0.0000
Operation-output L1	0.1676	0.0000
Frame-2 history switch	0.9961	0.0000
History-output L1	0.1421	0.0000
Minimum role localization	1.0000	0.0820
Frame-1 identity top-1	0.9961	0.3410 / 0.3197 / 0.2740
Frame-2 label top-1	0.9969	0.4873 / 0.3717

Table 7: V24 fresh paired development audit. The last two control cells list query/operation/header-blind and history/header-blind results, respectively. All arms have exactly 1,347 trainable parameters.

V24 frozen gate	Measured	Required	Result
Frame-1 binary choice	0.9980	> 0.95	pass
Query switch	0.9766	> 0.90	pass
Operation switch	0.9727	> 0.90	pass
Generated-history switch	0.9961	> 0.90	pass
Held-out minimum switch	0.9635	> 0.85	pass
Frame-1 identity top-1	0.9867	> 0.75	pass
Frame-2 label top-1	0.9973	> 0.95	pass
Frame-1 pixel F1	0.8371	> 0.68	pass
Frame-2 pixel F1	0.7286	> 0.58	pass
Generated-history consistency	0.9863	> 0.92	pass
Held-out $T = 24$, frame 1	0.9847	> 0.90	pass
Held-out $T = 24$, frame 2	1.0000	> 0.90	pass

Table 8: Single authorized V24 frozen evaluation. The sealed report records no model selection, no threshold change, and no repeated frozen run.

a hidden state. It does not establish arbitrary sentence parsing, factual retrieval, page continuation, historical form synthesis, free-form output length, or movie generation. V25 implements the next narrower question: remove packet algebra, train on ordinary rendered Chinese, and require the resulting visual history to beat causal and frequency controls before a writer is authorized.

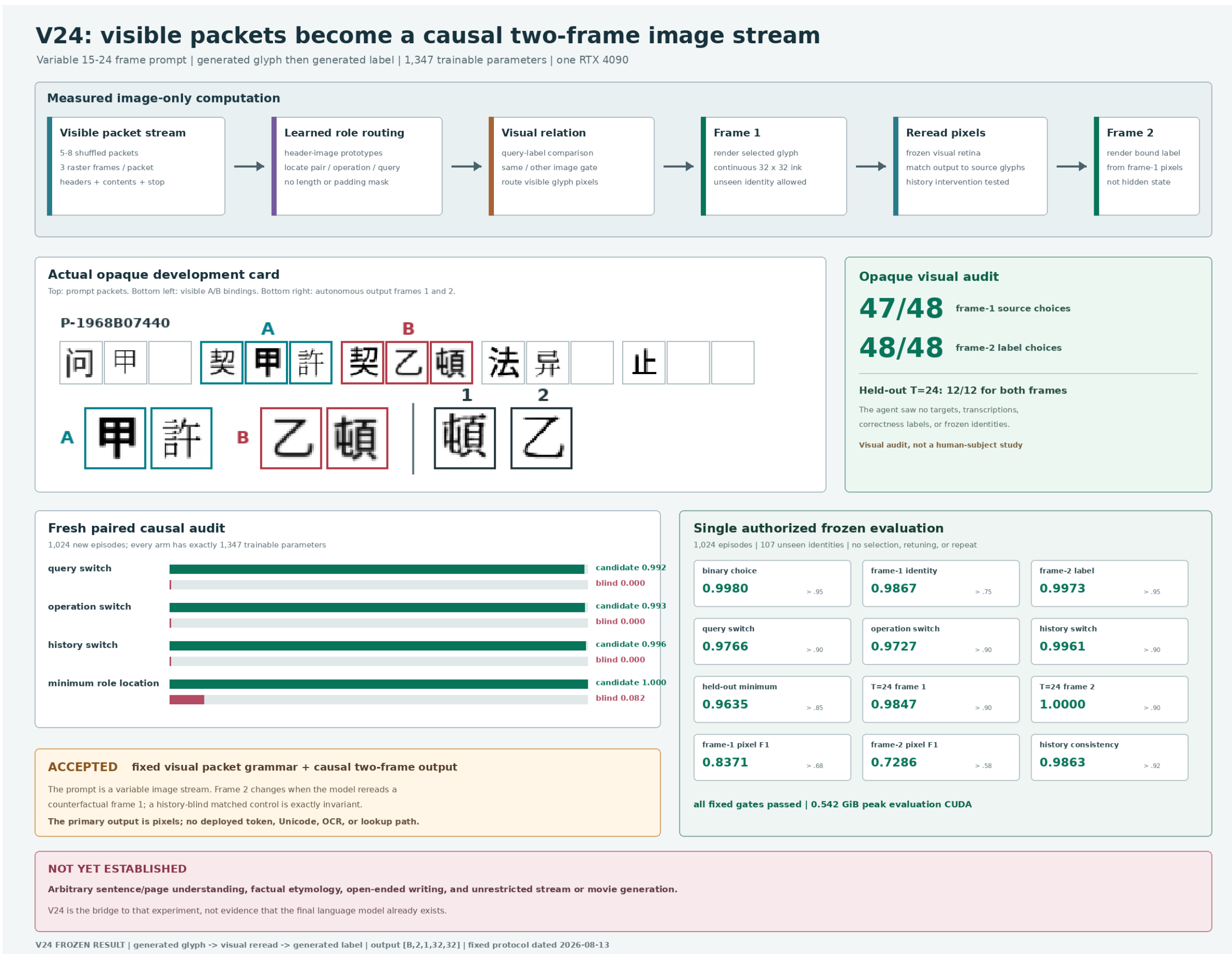

Figure 12: Measured V24 result. An actual opaque review card shows a variable-length image prompt and two autonomous answer frames. Matched controls localize header, query, operation, and generated-history causality; every gate in the single frozen evaluation passes.

5.8 Visual Cell Stream V25: natural Chinese prediction rejected

V25 treats each visible writing unit as an inspectable retinal observation and stacks observations along visual time:

$$X \in [0,1]^{B\times T\times 1\times 32\times 32}, \qquad T = 64.$$

The third nontrivial axis of one sample is reading order, not a character ID or geometric depth. The student receives only these grayscale images, their order, continuous corruption and flow times, random image noise, and its own generated pixels. Strings, bytes, token and Unicode IDs, OCR transcripts, character labels, candidate-bank rows, and external models are excluded from the learned path.

The fixed manifest contains 7,017 records from 16 public-domain Chinese works, with SHA-256 `76048753...f6866c03`. Hash partitioning by original record gives 6,608 training, 190 development, and 219 frozen groups. Training and development fonts are disjoint. Two independently rendered views of every 65-cell span reverse their context/target roles within a batch; their strings and renderer metadata are deleted before each model call. The frozen partition is counted but never rendered. An evaluator-only 1,024-form image bank and symbolic frequency tables are unavailable to the student and absent from deployment.

A frozen V16 retina maps each image to a continuous unit state $z_t \in \mathbb{S}^{191}$. Eight causal Transformer blocks of width 384 use rotary reading position and continuously corrupted prior images:

$$h_t = C_\theta(\tilde{z}_{\le t}), \qquad \hat{z}_{t+1} = \frac{P_\theta(h_t)}{\|P_\theta(h_t)\|_2}.$$

There is no character embedding or vocabulary projection. The fixed language objective combines independent-view cosine prediction with multi-positive in-batch visual contrast,

$$\mathcal{L}_{\text{language}} = 1 - \langle \hat{z}_{t+1}, z'_{t+1}\rangle + 0.5\mathcal{L}_{\text{visual contrast}}.$$

The complete model contains 25,549,714 parameters; 15,810,241 are trainable in the language stage. The run performs 2,400 BF16 AdamW updates with physical batch three, two view directions, and four-step accumulation, for 24 effective streams per update on one RTX 4090.

Development evaluation uses 2,048 fixed windows. The same model is queried with full history, only the last cell, shuffled prior history with the last cell fixed, and blank history. The evaluator separately computes image-unigram and symbolic-bigram controls from training frequencies. It also forms 512 pairs with the same last cell but different histories and different true next cells. Table 9 reports every selection-relevant measure.

Thus the causal field uses ordered history, but does not use it accurately enough to select the correct next form. Its top-1 changes on 77.93% of paired counterfactuals, yet changes to the correct target only 12.89% of the time. Context sensitivity alone is not understanding. The language stage is rejected, writer training is not authorized by the fixed path, and no frozen evaluation is opened.

For diagnosis only, a separate command explicitly marks itself exploratory and continues from the rejected language checkpoint. It freezes the causal field and trains the 7,073,617-parameter continuous flow writer for 1,200 updates. For clean target x, noise ϵ, and time τ it learns the velocity of $y_\tau = (1-\tau)x + \tau\epsilon$, conditions on h_t and $\hat{z}_{t+1}$, and rereads its sampled pixels. This run cannot change the evidence verdict. On 256 targets and 16 autonomous continuations, generated identity top-1 is zero, reread target cosine is 0.08015, and pixel F1 is 0.32210. Blank rate is zero and position-16 density is 0.97657 times target density. The writer therefore learns ink occupancy, not the intended continuation.

V25 development measure	Measured	Required	Result
Full-history top-1	0.01123	diagnostic	–
Last-only top-1	0.00146	diagnostic	–
Shuffled-history top-1	0.00342	diagnostic	–
Image-unigram top-1	0.01611	diagnostic	below
Symbolic-bigram top-1	0.12158	benchmark	below
Full minus last top-1	+0.00977	> 0.030	fail
Full minus unigram top-1	−0.00488	> 0.030	fail
Full minus last log probability	+0.04882	> 0.050	fail
Full minus shuffled top-1	+0.00781	> 0.015	fail
Counterfactual switch accuracy	0.12891	> 0.550	fail
Full-history target cosine	0.27513	> 0.550	fail
Student boundary / peak CUDA	clean / 0.598 GiB	clean / < 18	pass

Table 9: V25 fixed natural-Chinese development audit. Full history has a small advantage over last-only and shuffled history, but remains below the image unigram and symbolic bigram. Six semantic and causal gates fail.

V25 localizes the next problem to the predictive state rather than page geometry. Cross-font visual appearance, exact identity, and context-predictive meaning share one target state; cosine can improve toward a frequent visual centroid without selecting the next glyph. A subsequent preregistered model should separate an exact appearance field from a context-predictive residual and require both under matched state-shuffle interventions.

Long context has a separate exact representation. For a lattice width K, sequence index t maps serpentine-wise to

$$r = \lfloor t/K \rfloor, \qquad c = \begin{cases} t \bmod K, & r \text{ even}, \\ K - 1 - (t \bmod K), & r \text{ odd}. \end{cases}$$

This bijection preserves adjacency at row turns and maps clean $N \times 1 \times 32 \times 32$ cells to either an inspectable flat page or a learned $C \times R \times K$ retinal lattice with a validity mask. A 256×256 lattice holds 65,536 cells, but was not used in V25. Scaling this representation before repairing the 64-cell objective would increase context without establishing better language.

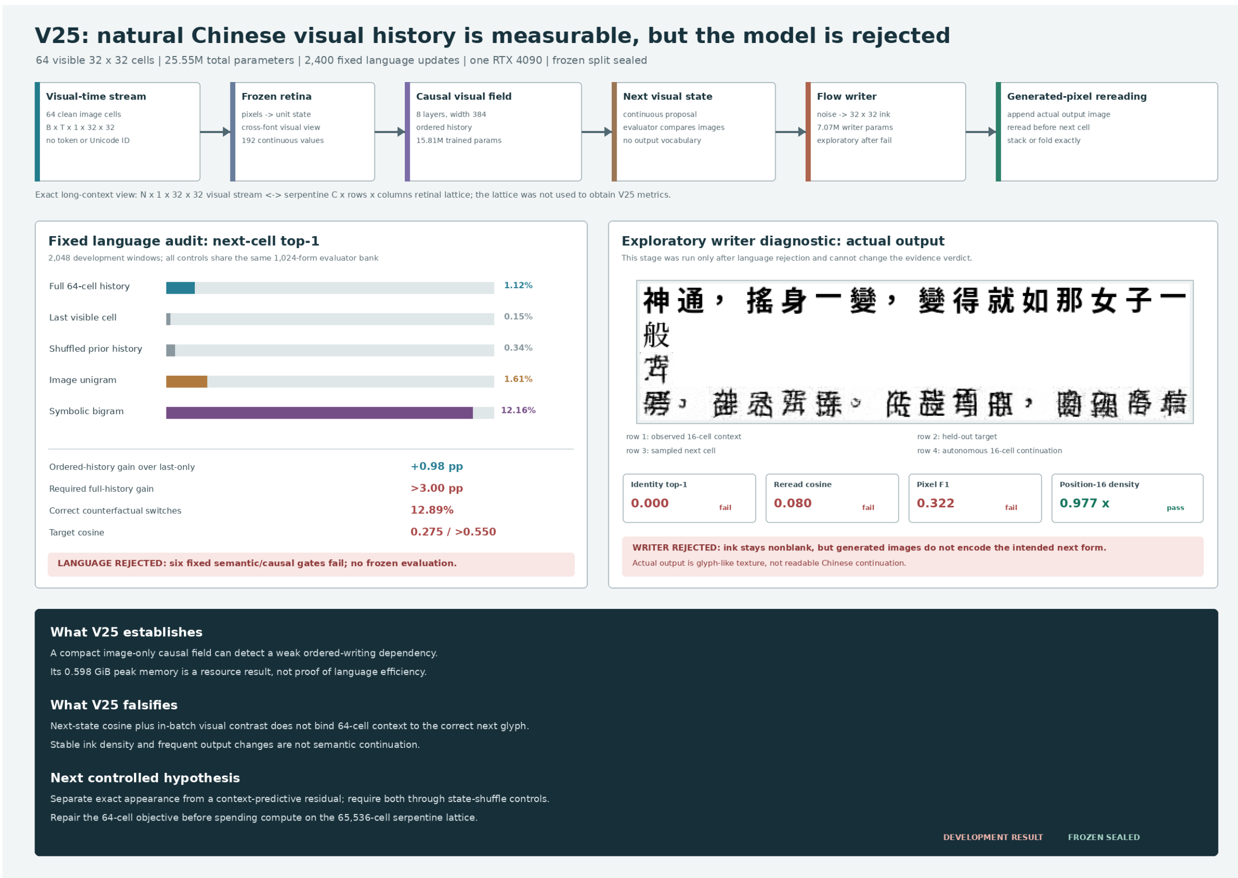

Figure 13: Measured V25 development result generated directly from the fixed language report, separately labeled exploratory writer report, and actual writer sample. Ordered visual history has a weak effect, but the unigram and bigram controls remain stronger and generated images are not the intended Chinese forms. The frozen partition stays sealed.

6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

For word-origin and book-layout research, 11 additional local works are registered by SHA-256 in `references/source_books/manifest.json`. They total 963,986,234 bytes and include 9,361 PDF pages plus one EPUB: English and Chinese encyclopedias, an English word-origin reference, and Chinese character- origin and oracle-form references. Relative symlinks avoid duplicating or committing the source binaries. Their rights are unverified, so they are private reference, source-comparison, and evaluation material only; they are not part of the releasable training corpus. PDF/OCR extraction may create offline sidecars, but those strings cannot enter a student batch.

7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t)\mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{\text{V7}} = \mathcal{L} + \kappa(t)\left(0.5\mathcal{L}_{\text{ctx}} + 0.2\mathcal{L}_{\text{sample}}\right).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [36, 35]; it remains image-only because the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal μ, target image state z^+, and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j\in\mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ \mid h_{\text{full}}) + \ell(z^+ \mid h_{\text{last}})\}.$$

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

8 Image-First Training and Inference

Figure 14 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 15 shows the user-facing path. The interface can behave like a normal LLM prompt box. Typed text is rasterized into a clean prompt band; an optional book page, inscription, or glyph crop is normalized and placed beside or below it on the same canvas. Borders, whitespace,

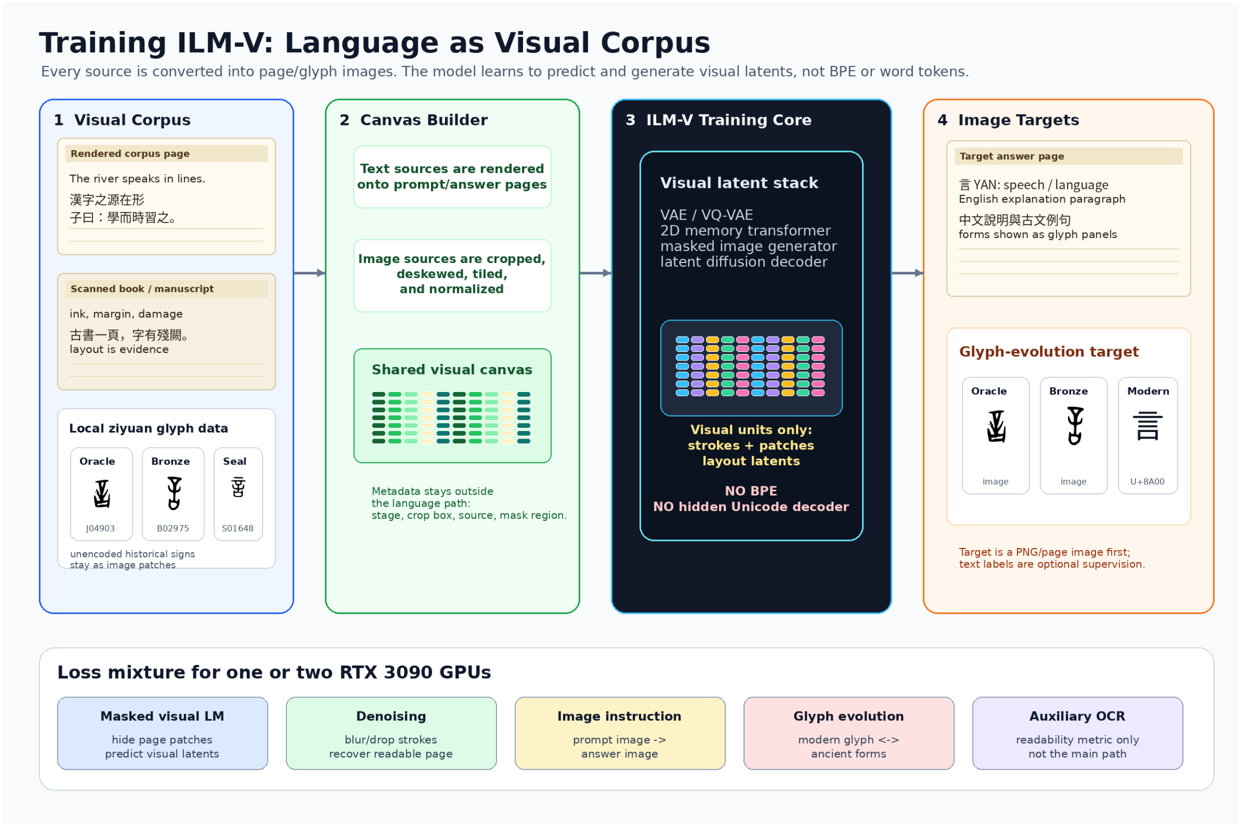

Figure 14: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

and coordinates make the relation visible rather than encoding it in hidden IDs. The ILM can thus answer a typed question alone or a grounded question such as "What does the marked form mean, and how did it evolve?" about an attached page. It returns a page image containing modern language and historical glyph regions. A later OCR/transcoder may attach a searchable layer for encodable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned}
r_\alpha(\text{text prompt}) &\to x_{\text{text}}, \\
g_\beta(\text{uploaded page or glyph}) &\to x_{\text{source}}, \\
\text{Compose}(x_{\text{text}}, x_{\text{source}}) &\to x_{\text{prompt}}, \\
f_\theta(x_{\text{prompt}}) &\to y_{\text{page}}, \\
o_\psi(y_{\text{page}}) &\to \text{optional text layer} + \text{glyph-region metadata}.
\end{aligned}$$

Here r_α and Compose are deterministic UI adapters, not language components. Either visual region may be absent. The core f_θ receives the resulting pixels and maps them to answer image y_{page}. The optional reader o_ψ may recover coded text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

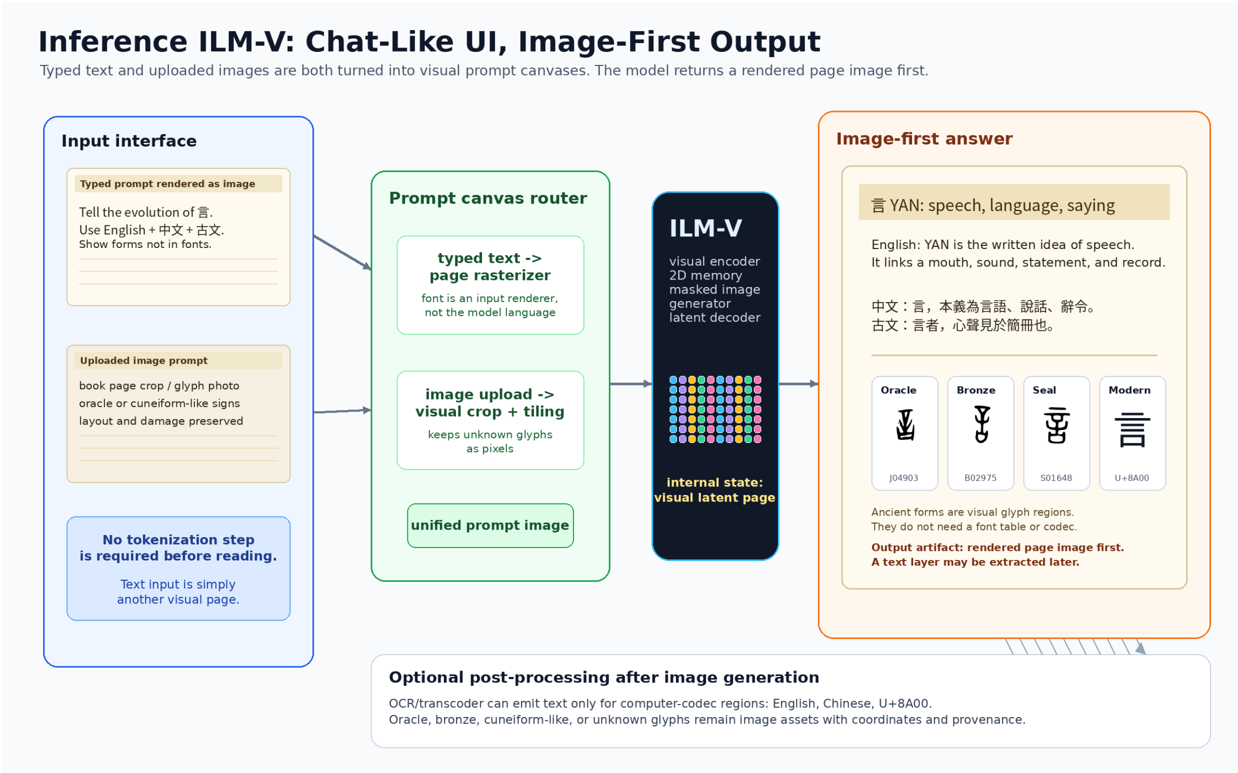

Figure 15: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from "text answer plus illustrations". The generated page image is the primary output object, and the text layer is secondary.

9 Single-4090 Curriculum

Figure 16 records the broad curriculum. The executable path is intentionally simpler than the complete research space:

1. Read 32×32 writing images into one global state and one small 4×4 spatial field. V16 already provides the retina and causal state.
2. Predict the next continuous visual state from image history and require it to beat last-only, unigram, and bigram controls.

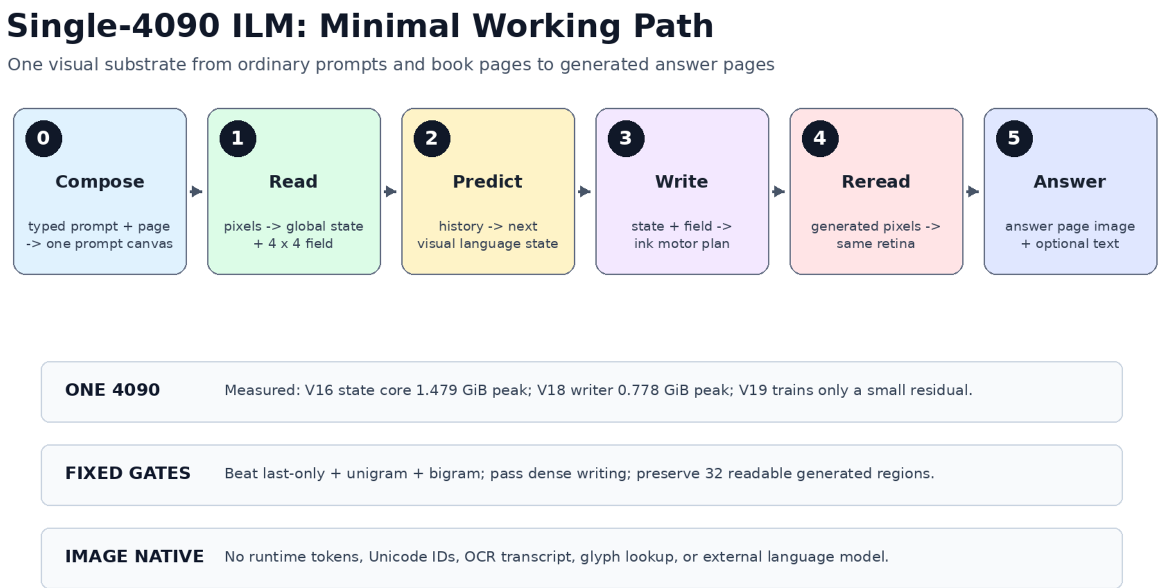

Figure 16: A staged curriculum that keeps each causal proof small enough for one RTX 4090 before any page-scale expansion.

3. Draw that state with a continuity-preserving local route. V19 shows that a small optional residual is ignored; V20 makes field detail causal; V21 makes both occupancy and detail field-causal but rejects disjoint patch rasterization. The successor may overlap only bounded neighboring patches.

4. Render typed prompts and optional source pages into visual frames, then predict local/global answer fields from open question–answer trajectories. Require held-out prompt counterfactuals and prompt-shuffle controls.

5. Feed generated answer pixels back through the same retina and require a readable, prompt-dependent multi-frame autonomous rollout.

6. Add model width or a second GPU only after the preceding fixed gate passes; use the second GPU for throughput or independent replication, not to hide an unresolved objective.

This uses one retina, one causal field, one deterministic motor planner, and one write–reread loop. Stochastic surface generation, billion-parameter scaling, and broad multimodal alignment are deferred until this minimal system works. The same Visual Language Stream can later expose geometric depth and time, but 3D glyph strings and character movies are also deferred until the 2D write–reread gate passes.

The 16.47M PVF V16 state model, 5.73M V17 actuator, and 2.36M V18 motor planner are deliberately small enough for one RTX 4090. Their measured peaks are 1.479, 1.588, and 0.778 GiB, respectively, leaving substantial capacity headroom for controlled corrections. V20 reduces the causal routing test to two exactly matched 0.506M-parameter arms, peaking at 0.323 and 0.397 GiB, and cheaply rejects the writer despite positive field causality. This is not an end-to-end efficiency

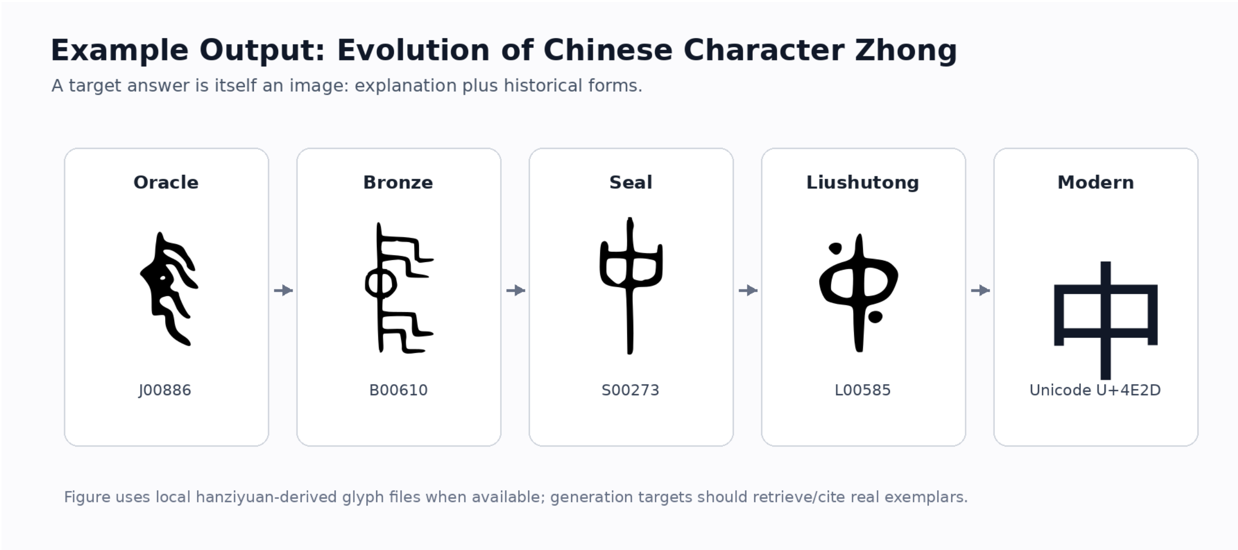

Figure 17: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

victory: the causal language core remains below a symbolic bigram, dense topology remains imperfect, and autonomous continuation has not passed. "More efficient than an LLM" remains an unsupported hypothesis until quality is matched on a named task.

10 Target Demonstration: Visual Word-Origin Book

The first demo should answer either a normal rendered prompt such as "Explain the evolution of the Chinese character yan" or the same question composed with a photographed book page. It must also support English word-origin questions. The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as "yan is the visible trace of speech",
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 17 is a compact stage-timeline example; the Yan figures in Figures 1 and 15 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

11 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

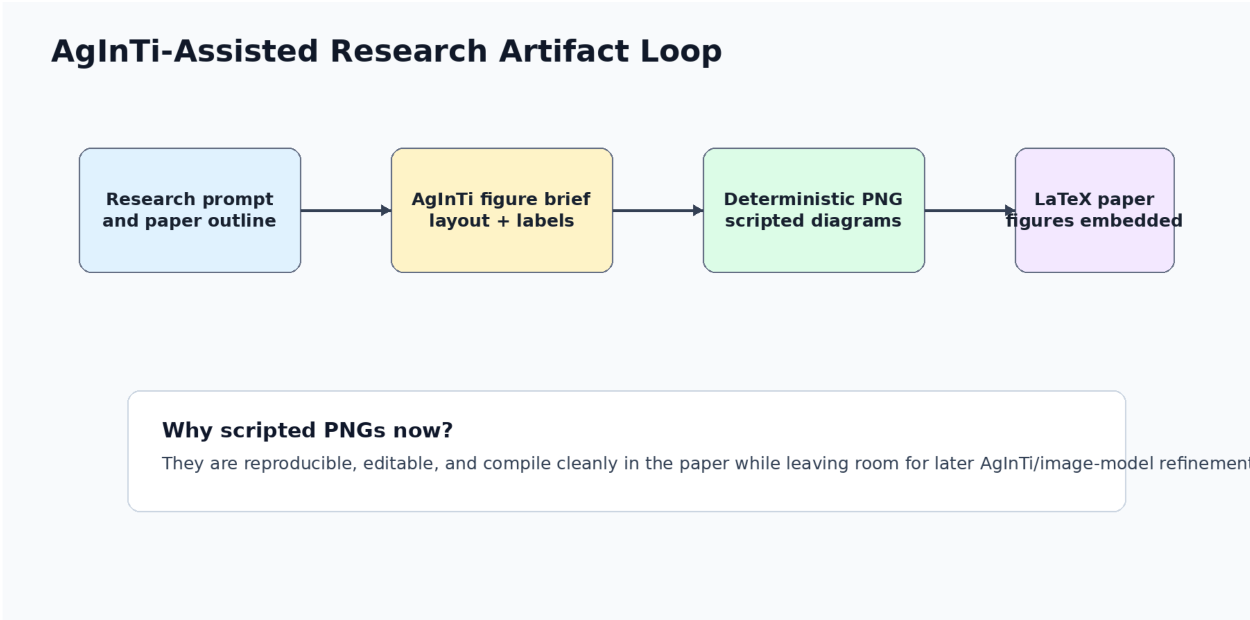

Figure 18: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

12 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Visual Word-Origin Book tasks: factual proposition accuracy, modern-text readability, prompt-image grounding, attested glyph-stage and source accuracy, unencoded-region preservation, and explicit synthetic-form labeling.
- Optional output transcription scored separately from the native answer image so an OCR sidecar cannot conceal a failed visual answer.

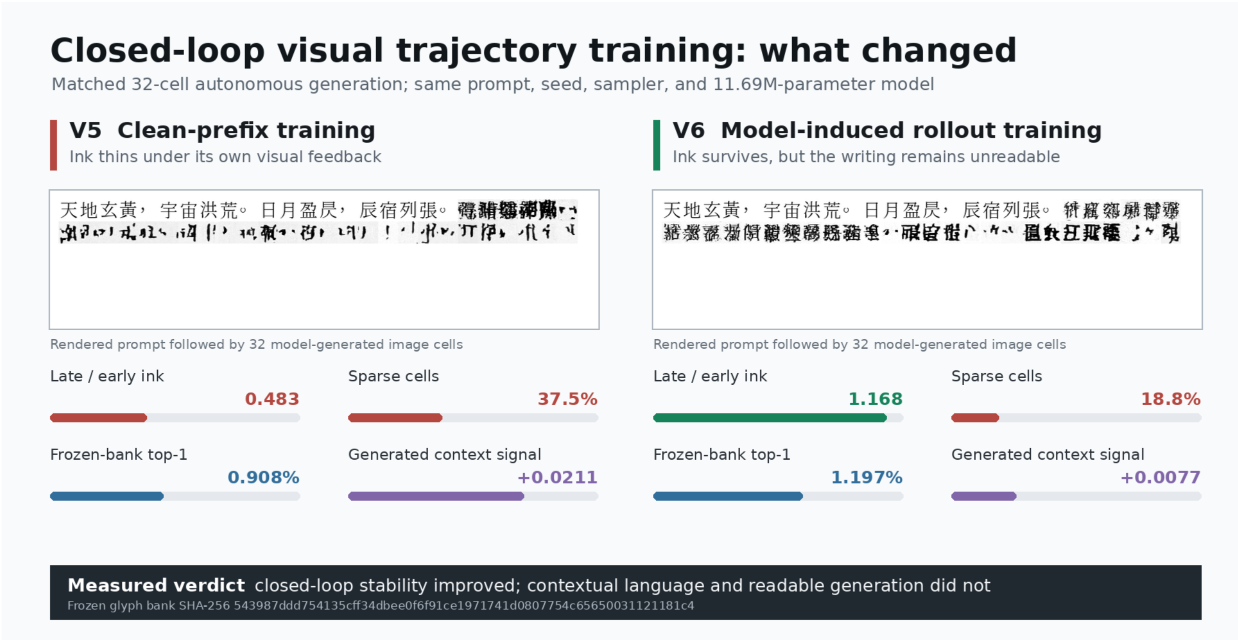

Figure 19: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain −0.2155	Much better than V6's −0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

Figure 20: Matched autonomous evidence for V6 and selected V7 step 5,800. V7 improves calibrated retrieval and generated target signal while retaining stable ink, but visual inspection still finds dense unreadable pseudo-glyphs.

The fixed evaluation contains 512 common Han candidates with four independent font views and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is -0.2155, improved from a corrected V6 value of -0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

PVF uses a separate frozen evaluation with the same 512 objects and four views for V14 through V16, bank SHA-256 `543987dd...1181c4`, and 1,788 eligible held-out contexts. V15 proposal top-1 is 5.872%, compared with 4.418% last-only, 1.734% unigram, 0.168% random dynamics, and 13.143% symbolic bigram. Proposal top-5 is 12.192%, and normalized context gain is $+0.07069$. The stochastic field obtains 3.412% top-1, 7.662% top-5, $+0.03032$ normalized context gain, and $+0.08053$ sampled context cosine gain. Random-dynamics sampled target cosine is 0.03180, versus 0.21576 for V15. The retina oracle is 98.546%.

V16 was selected at step 2,200 without reading that frozen bank and evaluated once. Its proposal obtains 6.264% top-1 (112/1,788), versus 3.971% last-only, 1.734% unigram, 0.224% random dynamics, 0.195% chance, and 13.143% symbolic bigram. Its proposal context gain is $+0.07732$. The stochastic field obtains 3.691% top-1 versus 3.244% last-only, normalized context gain $+0.03579$, sampled target cosine 0.19506, and sampled context cosine gain $+0.07947$. V16 adds seven correct proposal contexts over V15, while proposal top-5 and target cosine decline. This is directional evidence and not a statistically established memory advantage; paired predictions on a new frozen bank are required for attribution.

Thus the proposal and flow state gates are accepted, while the bigram language gate remains false. Separately, V17 accepts causal visual actuation and rejects readable topology. V18 then

PVF branch	Frozen signal	Interpretation
V14 proposal	top-1 2.52%	First proposal gate pass; above last-only 2.24% and unigram 1.73%
V14 state flow	top-1 2.35%	First stochastic state gate pass
V15 proposal	top-1 **5.87%**	Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707
V15 state flow	top-1 **3.41%**	Above last-only 2.68% and unigram; sampled context cosine gain +0.0805
V16 proposal	top-1 **6.26%**	112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773
V16 state flow	top-1 **3.69%**	Above last-only 3.24% and unigram; sampled context cosine gain +0.0795

Table 10: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

V17 frozen actuator metric	Correct state	Shuffled state	Gate
Global identity top-1	**58.594%**	0.977%	pass
Target cosine	**0.71301**	0.08612	pass
Pixel F1	**0.43849**	0.28001	fail (< 0.50)
Ink fraction	0.15947	0.15967	controlled
Human readability	rejected	rejected	fail

Table 11: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

passes every automatic development topology gate and makes many held-out forms readable, but leaves dense-form and prospective human-readability gates unresolved. V19 then rejects an additive local-field repair: dense F1 is 0.7278, but field shuffling reduces it by only 0.0088 and field removal by only 0.0054. V20 then forces unavailable within-block detail through the field. Correct dense F1 exceeds shuffled-field by 0.1218 and zero-field by 0.3613, with exact quadrant locality, so the local-causality hypothesis passes. Writer selection still fails, and endpoint dense advantage over the equal-capacity control is only 0.0179 against a fixed 0.03 margin. V21 removes the remaining global spatial route. At its best diagnostic snapshot, correct dense F1 exceeds shuffled-field by 0.1739 and zero-field by 0.3465, identity top-1 is 79.10%, target cosine is 0.8331, locality is 1.0, and every exact-basis invariant passes. The tiled-global control collapses to repeated patches. Overall, simple, and medium F1 remain below their fixed gates, so no candidate selects and no paired, human, or frozen stage is authorized. V22 then tests the first prompt-to-answer image stream. Its candidate changes answer identity correctly for only 0.78% of query-counterfactual pairs, versus 0% for the query-blind control, while identity and writer quality are nearly equal. Every candidate selector argmax lands on the operation image. Thus the implemented stream boundary works, but an entropy-sharpened single-frame selector cannot perform the required multi-frame relation. V23 then qualifies and freezes an image canonicalizer before learning the relation. Its candidate reaches 99.80% minimum query/operation switch on a fresh paired audit; the corresponding blind controls are exactly invariant. The opaque visual review and single frozen evaluation also pass, with 99.46% frozen identity top-1 and 0.7848 pixel F1. This turns V22's interface into the first accepted bounded image-prompt-to-image-answer circuit in the project. V24 then replaces six absolute frame roles with visible headers in a 15–24-frame packet stream and closes a two-frame

V18 development motor-plan metric	Correct intent	Shuffled intent	Gate
Global identity top-1	**73.633%**	0.977%	pass
Target cosine	**0.84618**	0.07162	pass
Pixel F1	**0.65772**	0.31290	pass (> 0.60)
Ink fraction	0.17505	0.17505	controlled
Human readability	simple/medium clear	unrelated	dense forms fail
Frozen images instantiated	no		sealed

Table 12: Fresh V18 development audit with style held fixed and only continuous intent shuffled. Automatic topology gates pass, but frozen promotion is withheld because the human rubric was not numerically prespecified.

visual autoregressive loop. The fresh candidate reaches 99.22% query, 99.32% operation, and 99.61% generated-history switch accuracy; each corresponding blind control is exactly invariant. The opaque audit scores 47/48 and 48/48 for the two output frames. On the single frozen run, frame-1 unseen-identity accuracy is 98.67%, frame-2 label accuracy is 99.73%, and held-out-length scores are 98.47% and 100%. Every fixed gate passes. V25 then removes the packet grammar and exposes the larger gap. Full natural- Chinese visual history reaches 1.123% top-1, above 0.146% last-only and 0.342% shuffled history, but below 1.611% image unigram and 12.158% symbolic bigram. Six fixed gates reject the language model. The exploratory writer's nonblank images preserve density while obtaining zero identity top-1, so appearance stability cannot substitute for semantic continuation. The frozen partition remains sealed. The combined result establishes that a small image-only student can learn causal language signal, that a small continuous field can control topographic generated ink, and that several visual roles can be composed into a rendered answer stream whose second frame depends on rereading the first. It does not establish useful autonomous language. V25 shows that simply replacing the hand-authored grammar with a generic causal visual field is insufficient. The next proof must first separate appearance preservation from context-predictive state at the same 64-cell horizon and pass matched state interventions. Only then should a serpentine lattice, variable answer length, or multiple generated lines expand the task. This order is also consistent with measured error accumulation in iterative diffusion chains [37].

13 Conclusion

The experiments show that merely replacing tokens with pixels is insufficient. A page flow can learn paper statistics, a causal writer can learn local ink, and a retina can learn character identity without acquiring language dynamics. RFLM contributes a strict test harness for ordered visual reading, direct ink flow, and autonomous rereading, but V6 and V7 reject exposure-bias correction and a monolithic language-plus-rendering writer as sufficient solutions. PVF then factors perception, causal language state, stochastic visual form, and actuation. V16's 6.26% proposal and 3.69% state-flow top-1, positive normalized history gains, and random/unigram/last-only controls replicate the accepted visual-state proof in this project. They show that causal language signal can be learned directly from writing images by a 16.47M model on one consumer GPU. V17 then reaches 58.59% frozen visual-identity retrieval versus 0.98% under shuffled intended states with only 5.73M trainable parameters and 1.588 GiB peak memory. This breaks the categorical claim that lightweight continuous visual intent cannot control generated writing. Its 0.438 pixel F1 and unreadable pseudo-characters reject the stronger claim. V18 then uses only 2.36M trainable parameters and 0.778 GiB peak memory to reach 0.658 pixel F1 and 73.63% identity top-1 on fresh development renderings, versus 0.313 and 0.98% when intent is shuffled. This demonstrates that structured, recognizable writing can be generated as continuous image topology by a compact model on one consumer GPU without token unembedding. Dense forms still fail, the V18 frozen bank remains sealed, and the intended state is supplied rather than predicted. V19 adds a clean split, fixed complexity strata, five interventions, and a prospective blinded rule, then rejects its own proposed repair: the correct field gains only 0.0088 dense F1 over a shuffled field and 0.0054 over no field. This demonstrates that exposing local features is not equivalent to using them causally. V20 changes the routing constraint with two exactly matched 0.506M-parameter arms. It obtains a 0.1218 dense-F1 field-shuffle gap, a 0.3613 zero-field gap, and 1.0 local-occlusion locality. This establishes necessary, topographic use of the continuous retinal field. V20 is nevertheless rejected as a writer: overall F1, target cosine, exact decomposition, and the paired control margin fail, so no checkpoint, human review, or frozen evaluation is accepted. V21 then makes both coarse occupancy and detail originate from each corresponding local cell, with global state restricted to uniform modulation. Its 0.1739 shuffled-field dense gap, 0.3465 zero-field gap, 79.10% identity, 0.8331 target cosine, 1.0 locality, and exact basis receipts establish the stronger field-complete routing result. The equal-parameter global-only source cannot draw a spatial plan. Yet 0.6038 overall, 0.5648 simple, and 0.5945 medium F1 reject V21's disjoint-patch writer. No candidate selects, and no paired, human, or frozen claim is accepted. V22 then closes the interface for a bounded six-frame visual prompt and one-frame image answer with two exactly matched 3.410M-parameter arms. It rejects the stronger understanding claim: 0.0078 counterfactual switch accuracy, 0.0089 paired-output L1, and 0.1592 identity top-1 are not meaningfully better than the query-blind control's 0, 0, and 0.1533. The operation frame receives all 1,024 candidate selector argmaxes. This demonstrates why prompt-dependent image generation requires relation composition, not simply a visual Transformer, sharp attention, and a pixel writer. V23 supplies that composition with a separately qualified writer and three matched 25,602-parameter relation arms. Fresh development controls, an opaque visual review, and the single frozen run agree: the candidate uses both query and operation images, preserves pair-order invariance, generalizes to held-out compositions and 98 unseen identities, and renders the selected form. Frozen binary choice is 0.9983, identity top-1 is 0.9946, and pixel F1 is 0.7848. The maximum supported claim is fixed-grammar visual relation following. The roles and relation algebra remain built into the circuit, and the answer remains a canonicalized visible source form. V24 removes absolute role positions, varies the prompt from 15 to 24 raster frames, and emits two images. Five matched 1,347-parameter arms localize the causal chain: query-, operation-, and history-blind controls each have exactly zero response to their hidden factor, while header blindness destroys role localization. The candidate's single frozen run over 107 unseen identities reaches 0.9867 frame-1 identity, 0.9973 frame-2 label accuracy, and 0.9961 history switch accuracy; every fixed gate passes. V24 therefore establishes generated-pixel rereading in a bounded variable-input stream. It does not turn the fixed packet algebra into natural language. V25 performs that first ordinary-language test at a bounded 64-cell horizon. Its 1.123% full-history top-1 exceeds last-only by 0.977 percentage points and shuffled history by 0.781 points, establishing a weak ordered dependency. It does not exceed image unigram, misses the log-probability gate by 0.00118 nat, fails counterfactual and target-cosine gates, and remains 10.8 times below the symbolic bigram. The exploratory writer reaches zero generated identity. The 25.55M-parameter system fits easily on one RTX 4090, but capability-normalized efficiency is not established. The frozen partition is not queried. The seven-context V16 increase and V25 ordered-history increase are therefore both too small to establish a useful causal memory core. The next attributable test must keep ordinary Chinese and the strict image boundary while separating exact appearance from a context-predictive residual at the 64-cell horizon. Matched appearance-shuffle, predictive-state-shuffle, last-only, order-shuffled, visual-unigram, and symbolic-bigram controls must pass before long context or a writer is promoted. The exact serpentine mapping may then fold clean 32×32 cells into a 256×256 retinal lattice without making an 8×8 thumbnail the sole record. Page-native canvases, 3D geometry, color, and motion remain later observable channels and may not encode hidden character IDs. Prompt-paraphrase, fact-substitution, evidence-removal, generated-history-blind, retrieval, and OCR-plus-text controls must distinguish language use from reconstruction and lookup. Only selected components may enter longer visual memory. The named product gate is a Visual Word-Origin Book that reads a rendered prompt alone or composed with a source page, then emits one answer image or an ordered answer-image stream plus an optional declared text sidecar. Prompt counterfactuals must change that answer correctly; input reconstruction and glyph retrieval do not constitute understanding. Historical etymology answers and broad instruction following remain downstream goals, not present capabilities, and no Qwen-8B parity or end-to-end efficiency claim is warranted. The perception–prediction–motor–self-perception factorization may later use the Visual Language Stream for 3D character strings or writing movies, and may be tested on continuous speech or vocal fields. None of those higher-dimensional sensory-language extensions is established here.

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